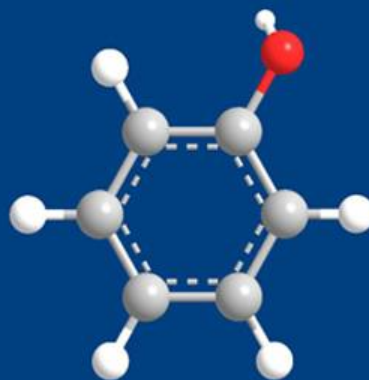
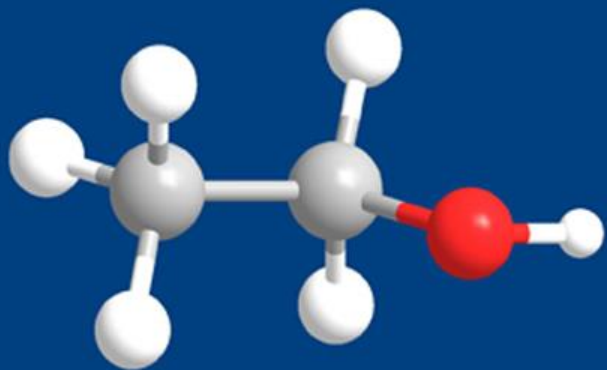




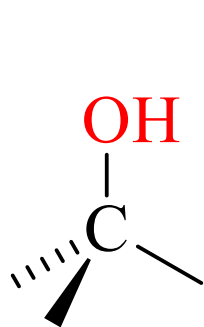
第九章 醇、酚和醚

Alcohols, Phenols and Ethers

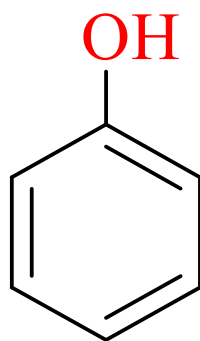


醇的分类、结构和命名

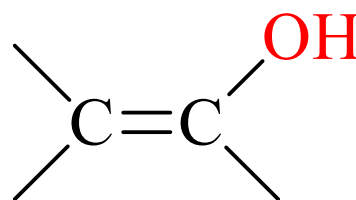
醇是羟基连接在饱和 sp^3 碳原子上的化合物



醇



酚



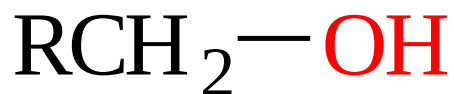
烯醇（不稳定）
（在醛酮中讲解）

- 醇可看成是水分子中的一个氢原子被烃基取代所形成的有机衍生物
- 醇的官能团是羟基(hydroxyl group) —OH

醇的分类

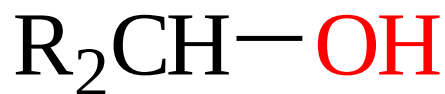
3种分类方法：

1) 根据官能团所连烃基类型



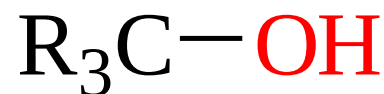
伯醇

一级醇(1°)



仲醇

二级醇(2°)



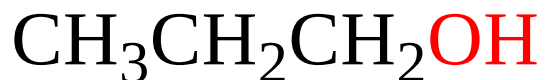
叔醇

三级醇(3°)

醇的分类

2) 根据烃基结构

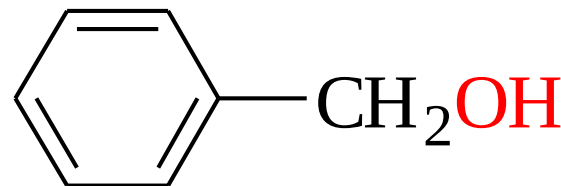
饱和醇



不饱和醇



芳香醇

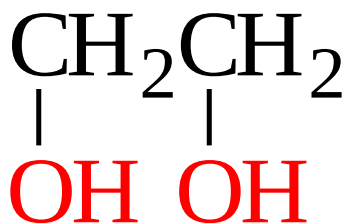


3) 根据羟基数目

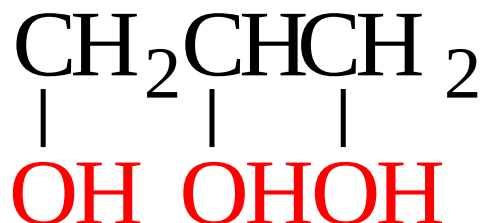
一元醇



二元醇

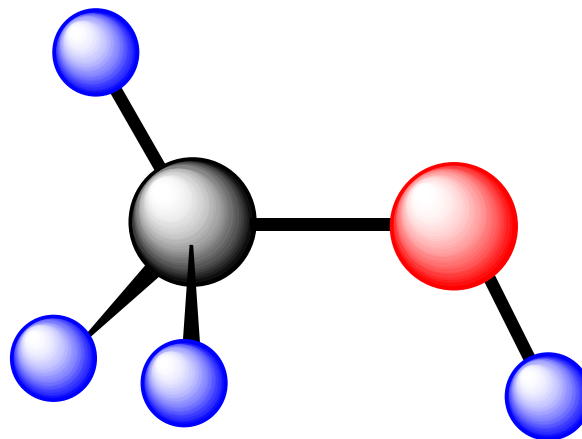
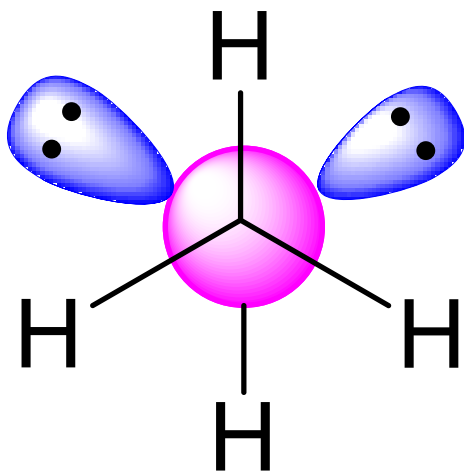


多元醇



醇的结构

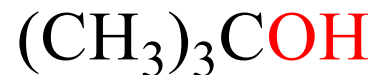
C、O都是 sp^3 杂化



正丁醇



异丁醇

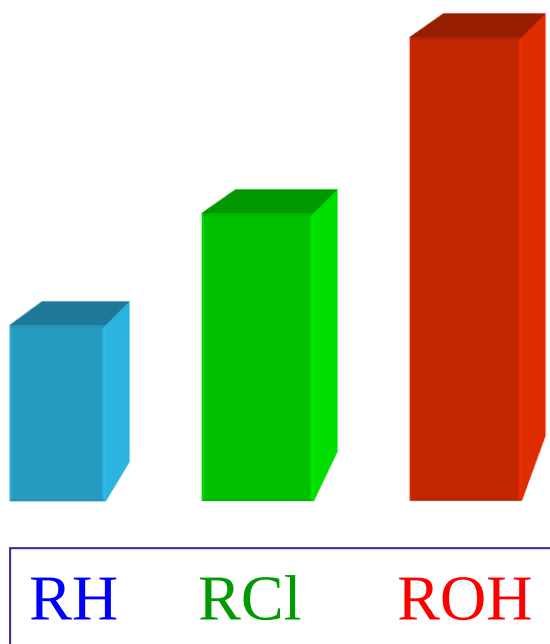


叔丁醇

醇的物理性质

● 醇的沸点

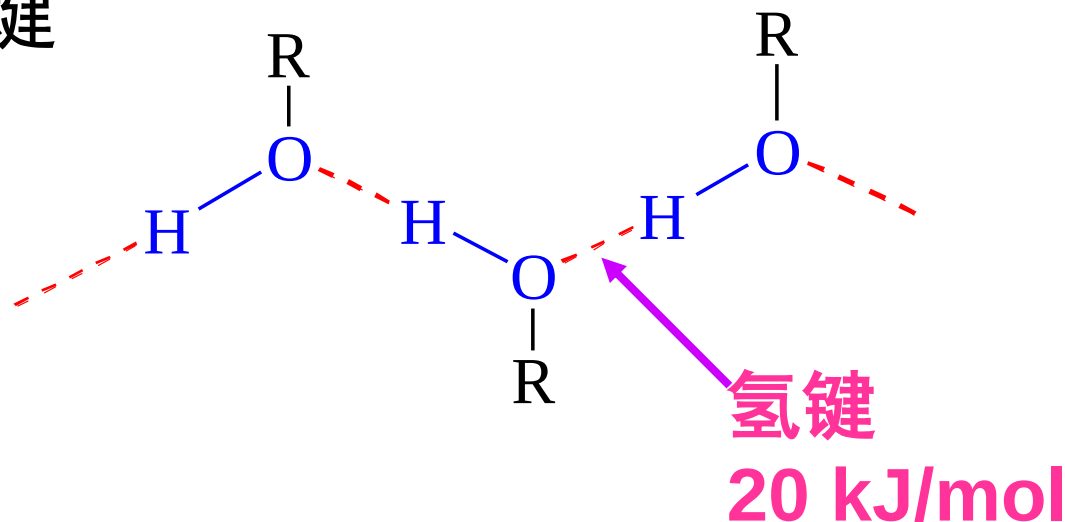
比同碳原子数的烷烃、卤代烃高，也比分子量相近的烷烃高



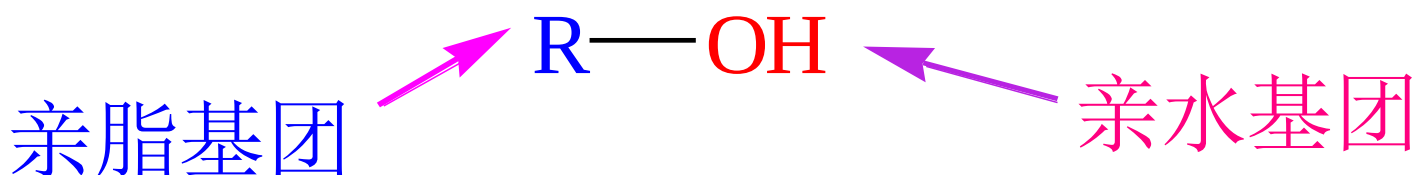
化合物	分子量	沸点(°C)
CH ₃ CH ₂ OH	46	78.5
CH ₃ CH ₃	30	—98
CH ₃ CH ₂ CH ₃	44	—42
CH ₃ CH ₂ Cl	64.5	12

醇的物理性质

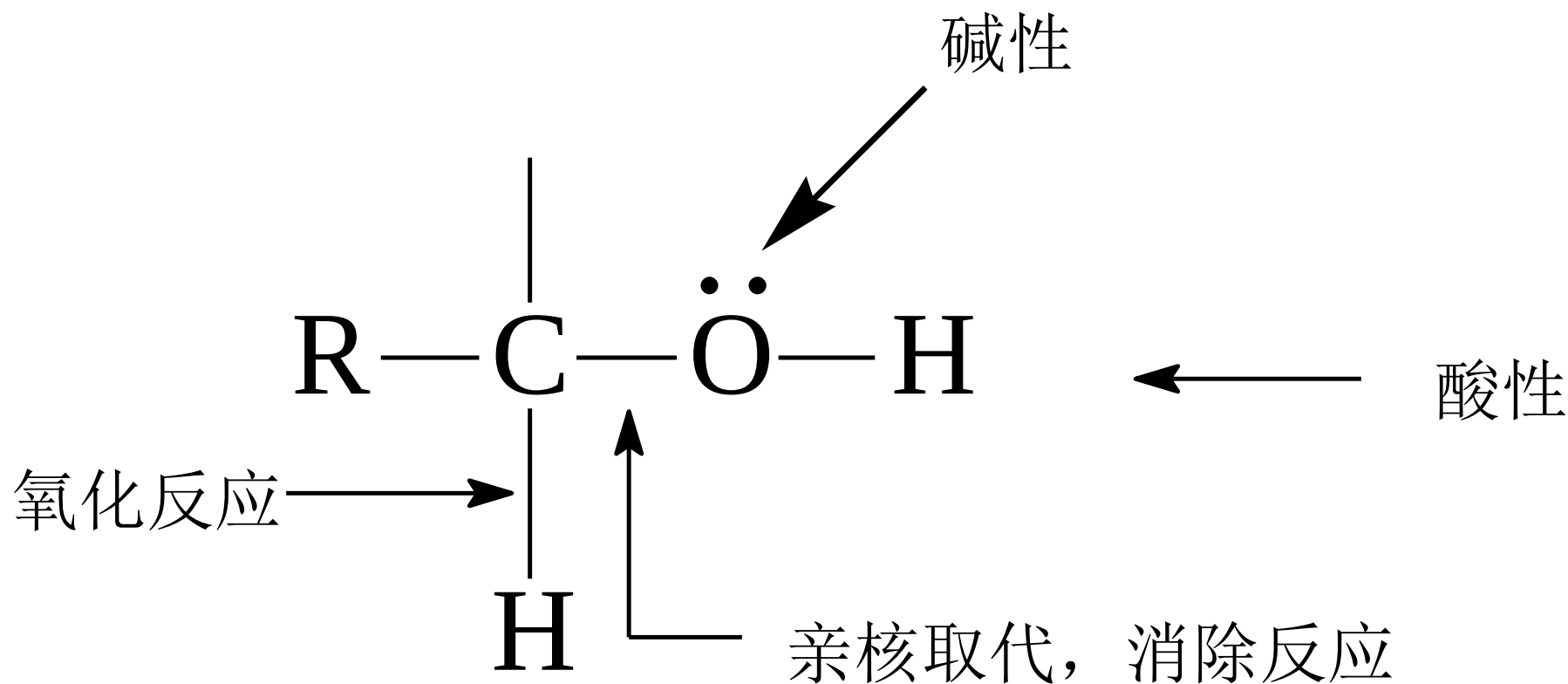
- 醇的分子间氢键



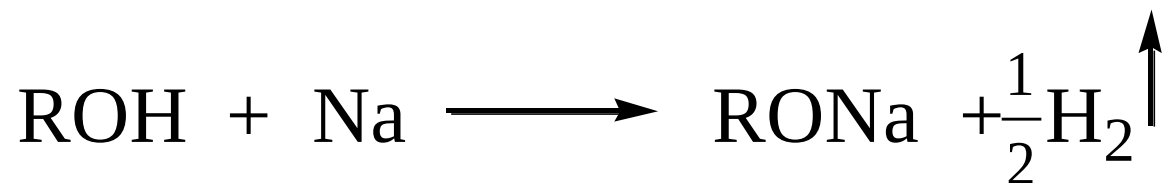
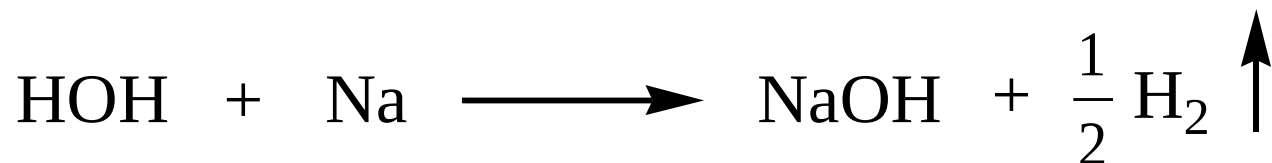
- 溶解度：低级醇溶于水，随分子量增加溶解度降低



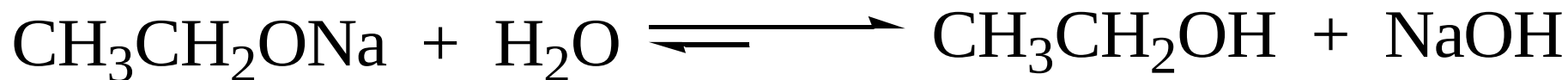
醇的化学性质



醇的酸性

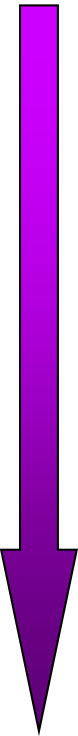


反应速度 $\text{CH}_3\text{OH} > \text{CH}_3\text{CH}_2\text{OH} > \text{CH}_3\text{CH}_2\text{CH}_2\text{OH} > (\text{CH}_3)_2\text{CHOH} > (\text{CH}_3)_3\text{COH}$

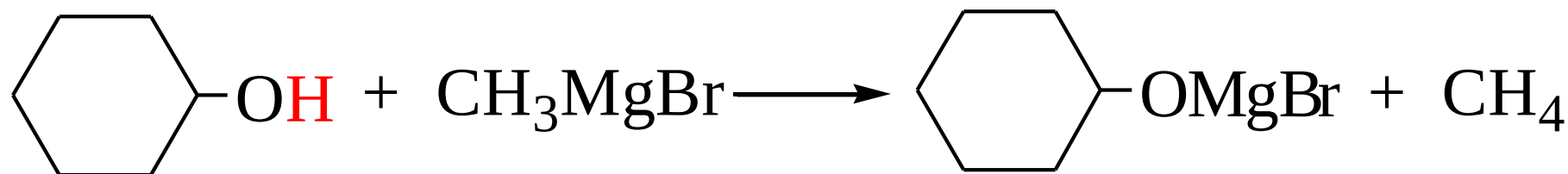
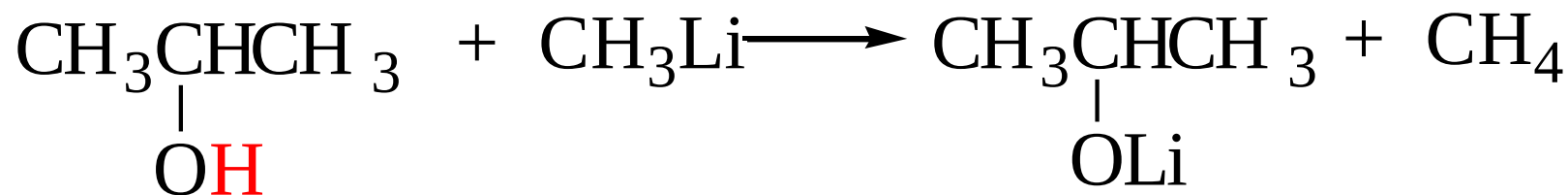
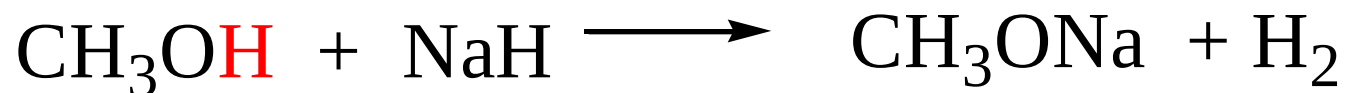


醇是一种弱酸

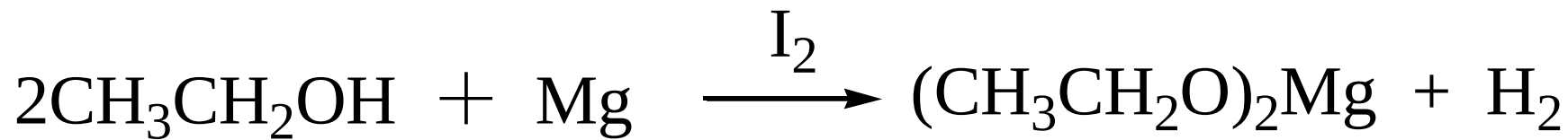
酸性大小

化合物	pK _a	
(CH ₃) ₃ COH	18.00	弱酸
CH ₃ CH ₂ OH	16.00	
HOH	15.74	
CH ₃ OH	15.54	
CF ₃ CH ₂ OH	12.43	
(CF ₃) ₃ COH	5.4	
HCl	—7.00	强酸

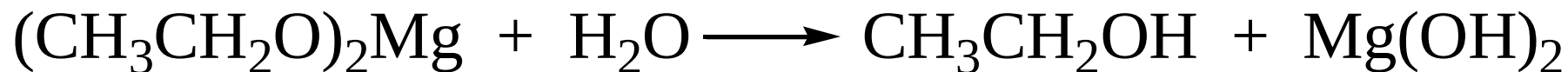
Example



醇与Mg反应



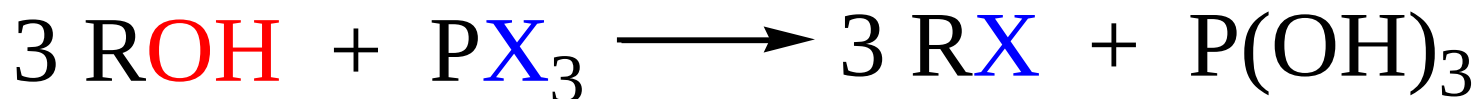
乙醇镁



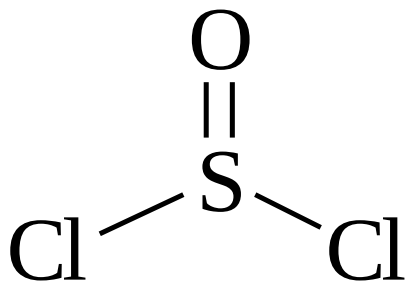
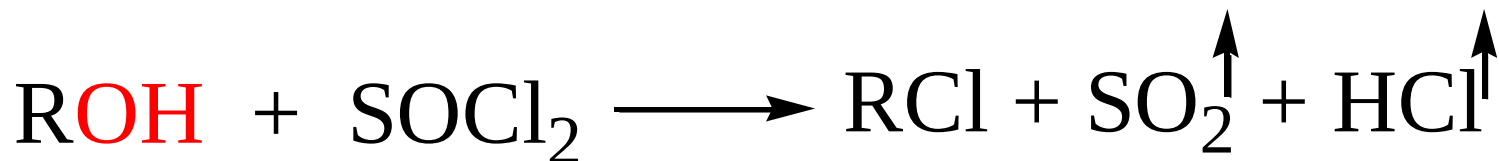
用于制备高度无水乙醇

取代反应

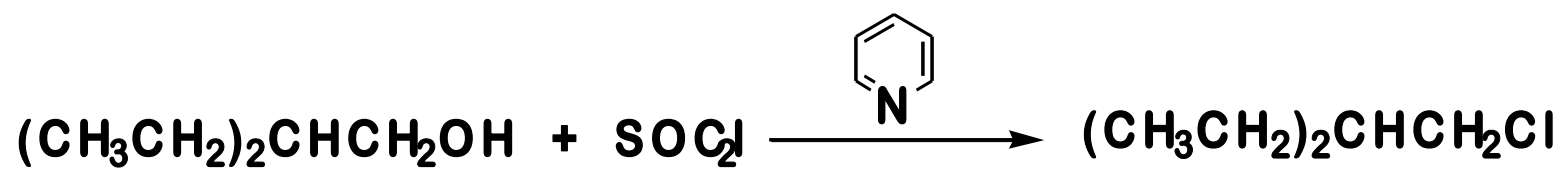
1. 与三卤化磷和氯化亚砷反应



或 PX_5 或 X_2 , P $\text{X} = \text{Br}, \text{I}$

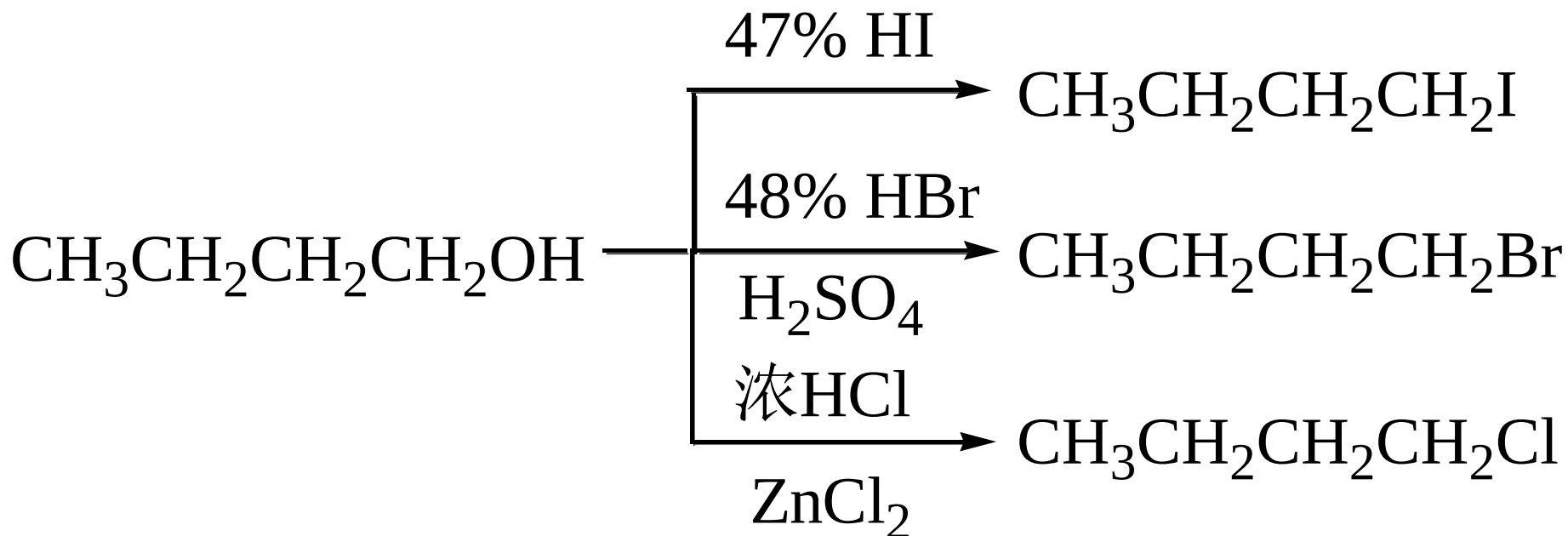


氯化亚砷
亚硫酸氯



取代反应

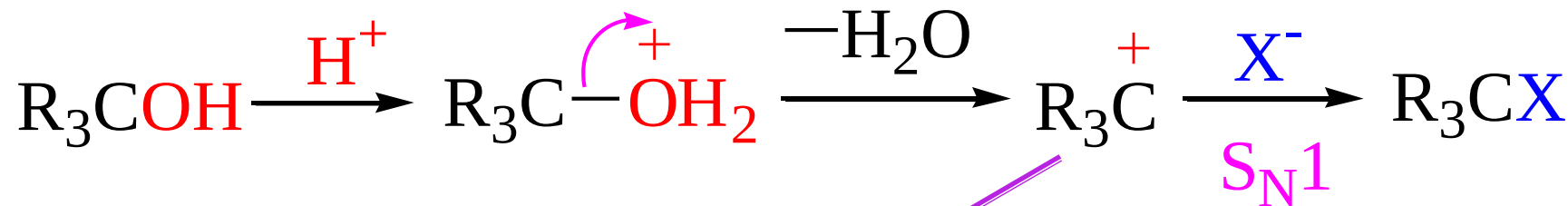
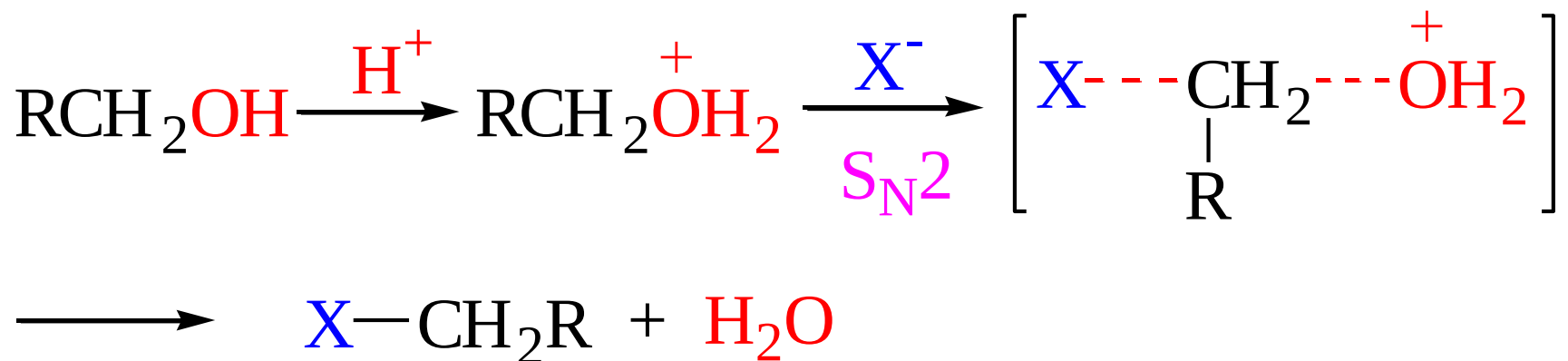
2. 与氢卤酸反应



反应活性: $\text{HI} > \text{HBr} > \text{HCl}$

$3^\circ \text{ 醇} > 2^\circ \text{ 醇} > 1^\circ \text{ 醇}$

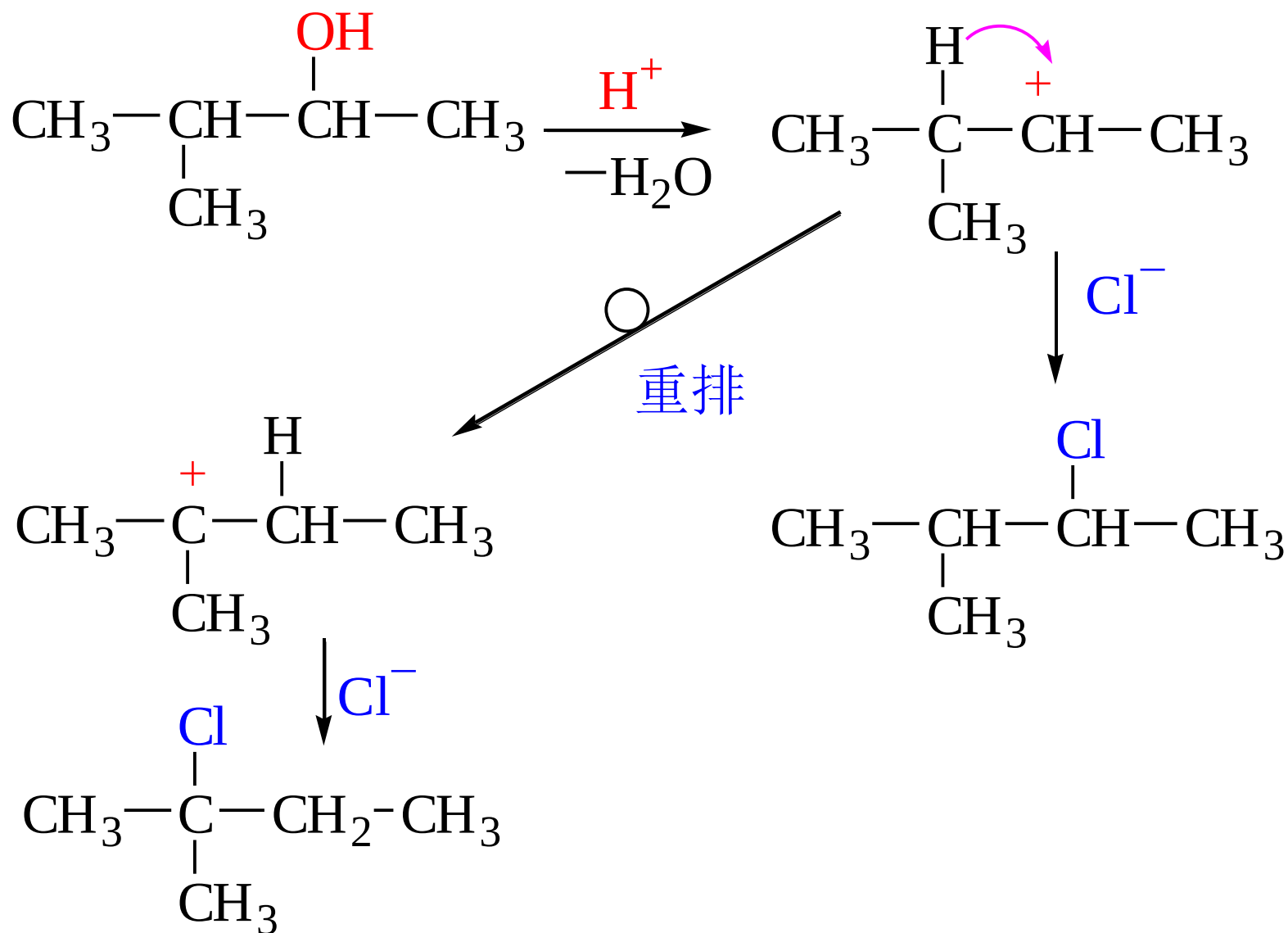
反应机理



仲醇: $\text{S}_{\text{N}}1, \text{S}_{\text{N}}2$

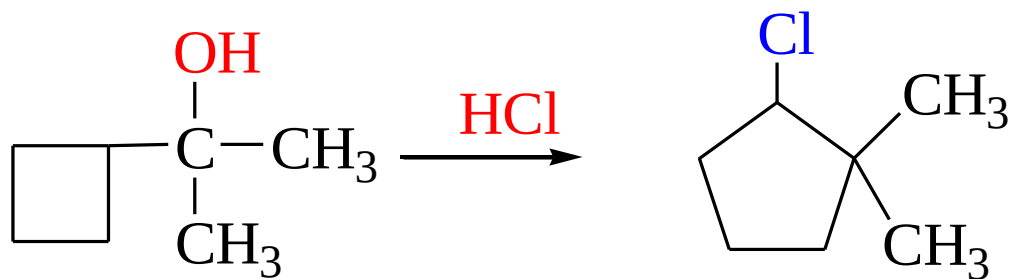
重排, 消除.....

反应机理

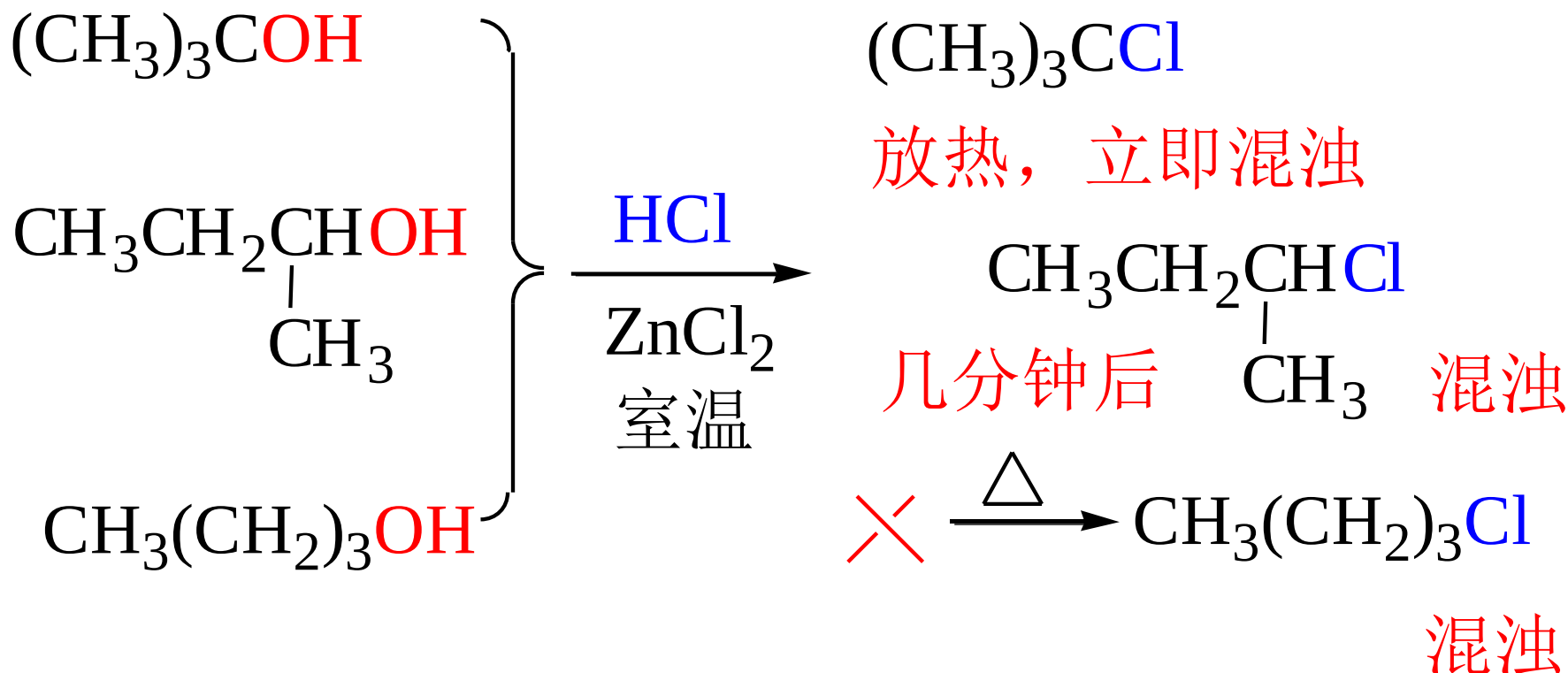


测试

请同学们用机理解释如下反应



Lucas Reagent (浓HCl/ZnCl₂)

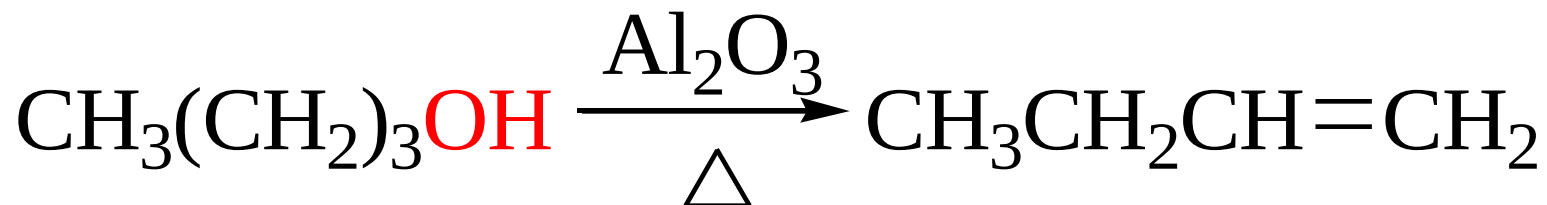


用于伯、仲、叔醇的鉴别
适用于C₄ ~ C₈的醇

脱水反应

常用催化剂：H₂SO₄, H₃PO₄, Al₂O₃

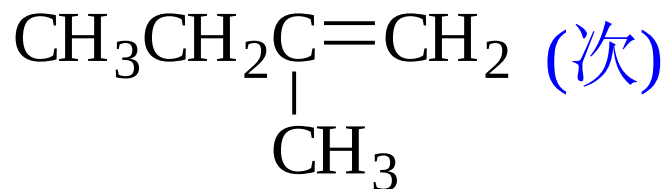
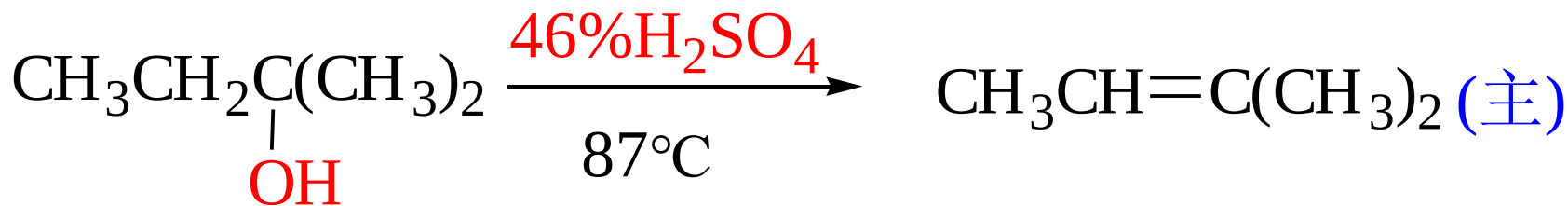
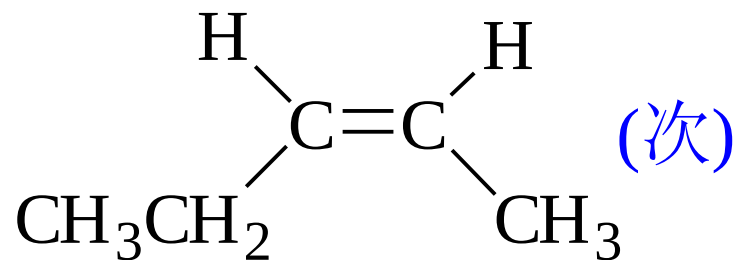
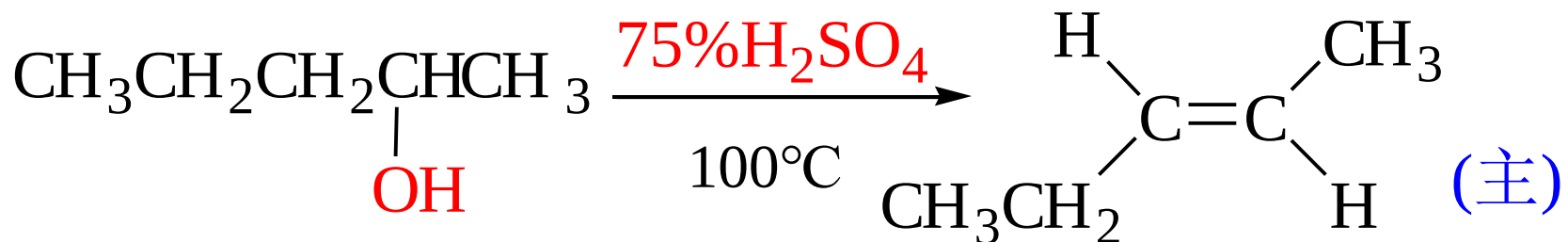
1. 分子内脱水



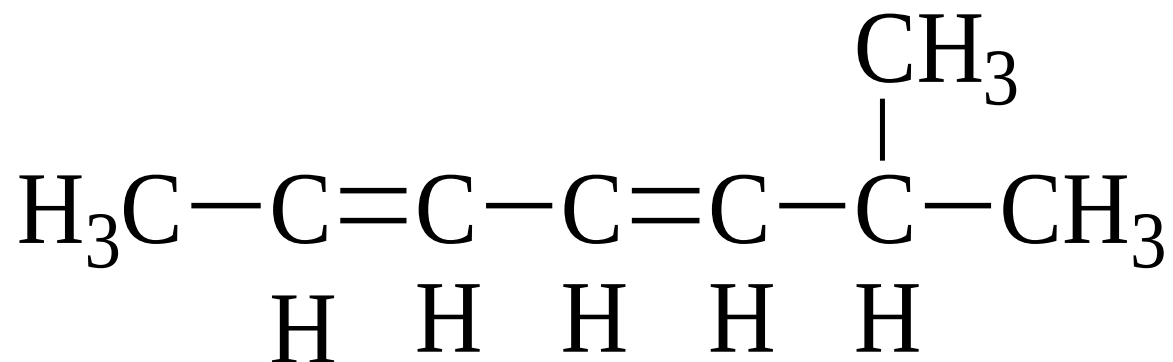
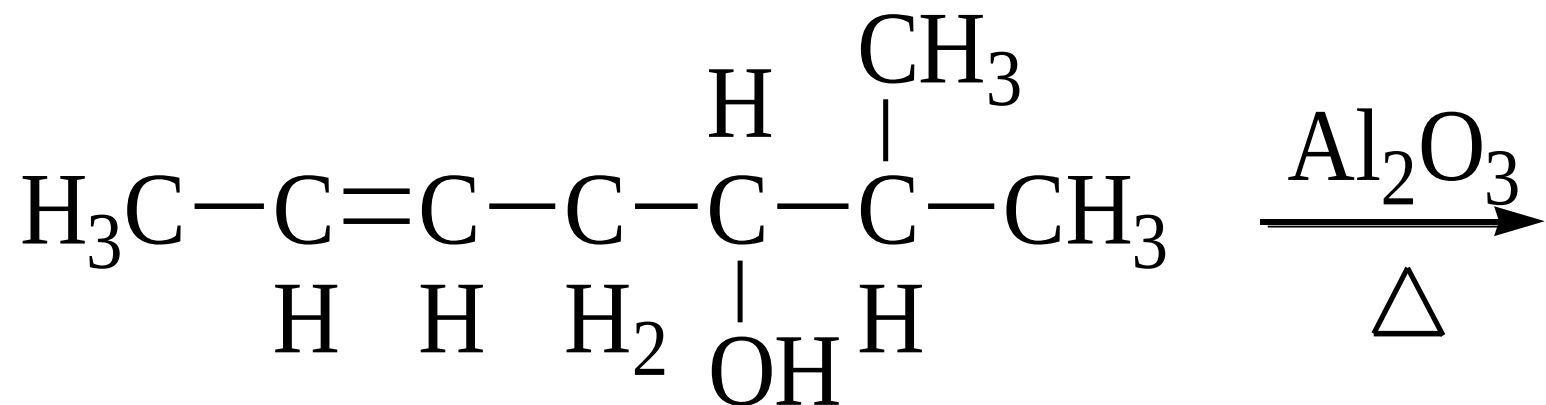
反应活性：R₃COH > R₂CHOH > RCH₂OH

叔醇的脱水较常用，有时也用仲醇，伯醇少用

机理：E1



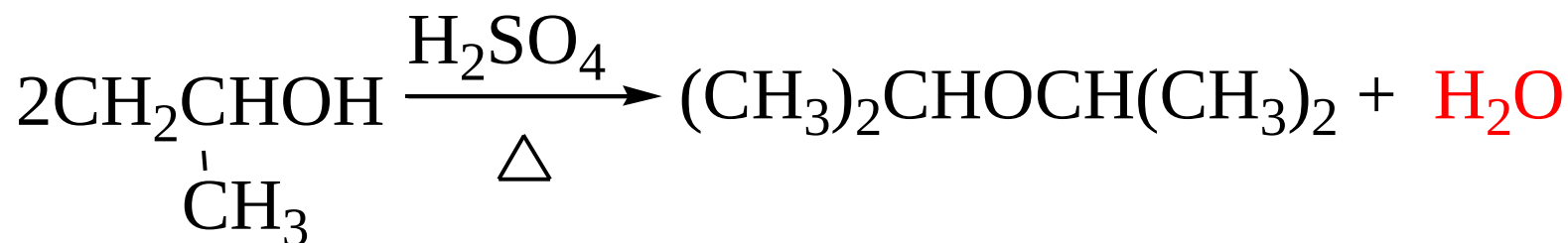
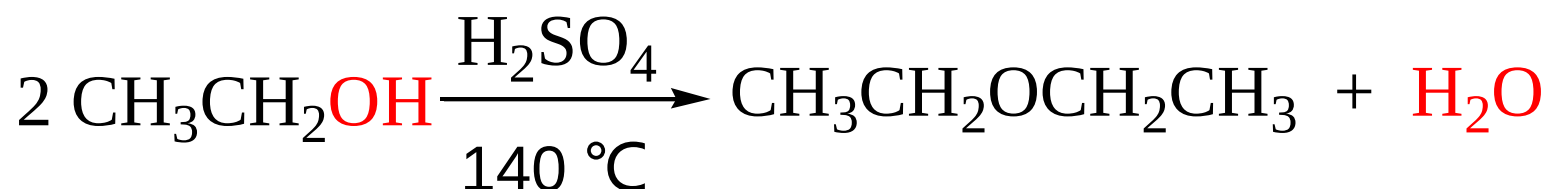
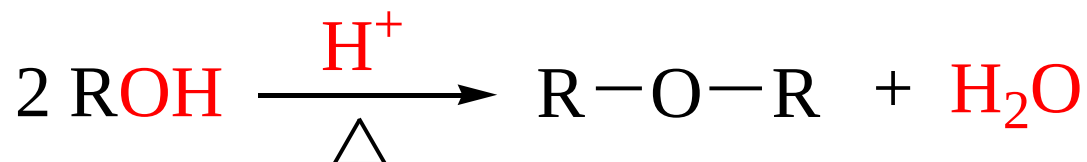
遵从Zaitsev 规律



共轭规律优先于Zaitsev规律

脱水反应

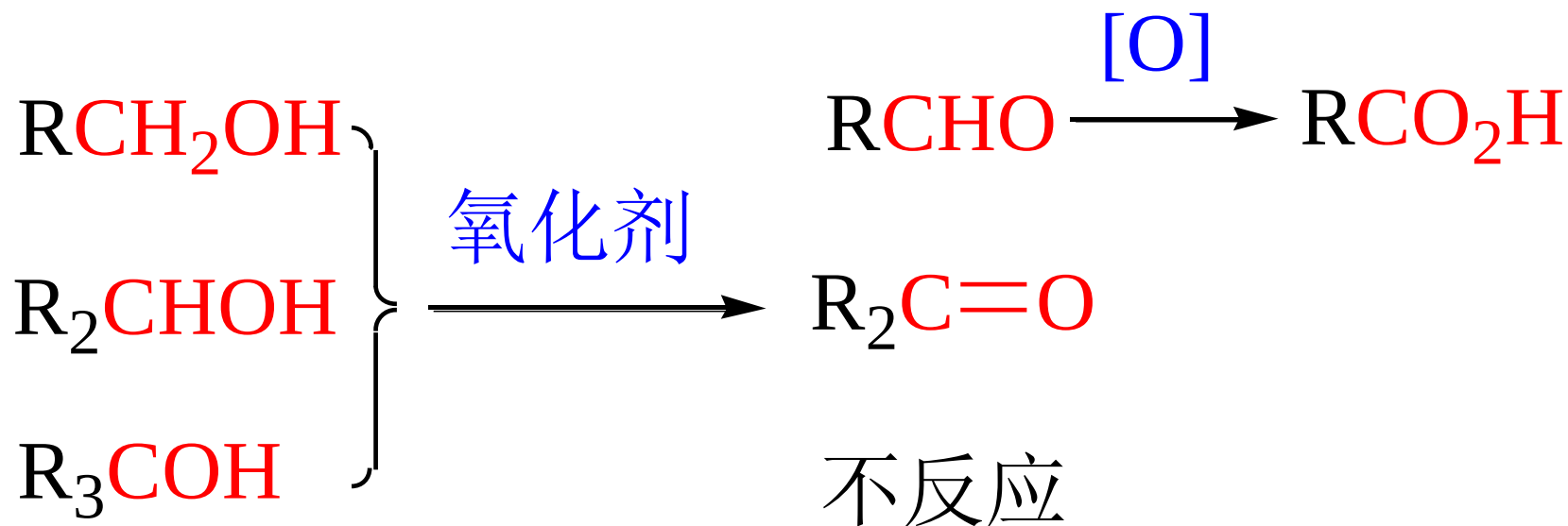
2. 分子间脱水



主要副反应：分子内脱水。
不适用于三级醇：空间位阻大

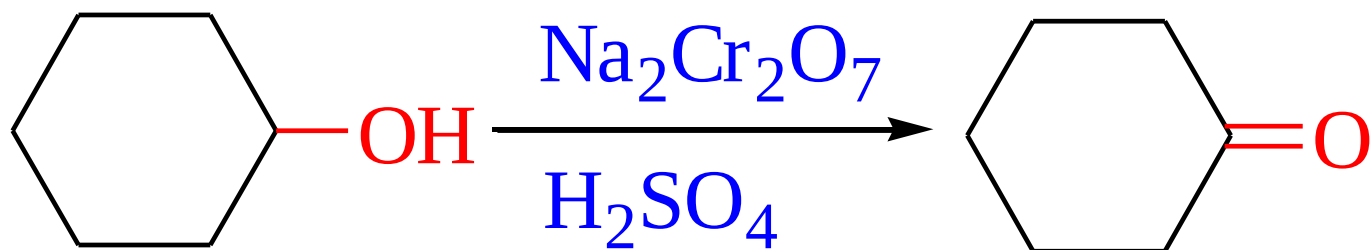
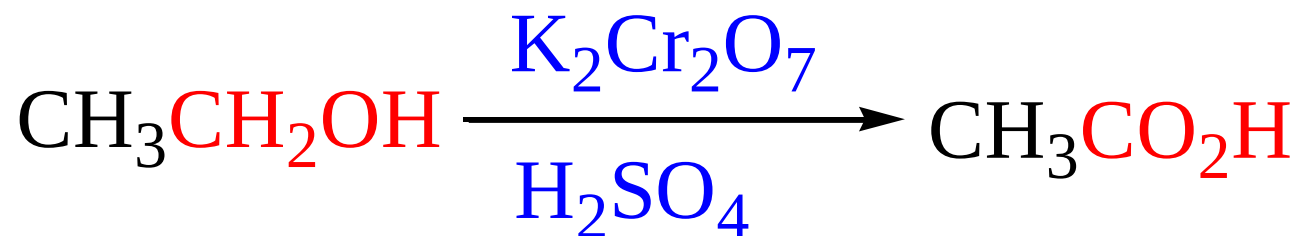
氧化与脱氢

- 氧化剂氧化

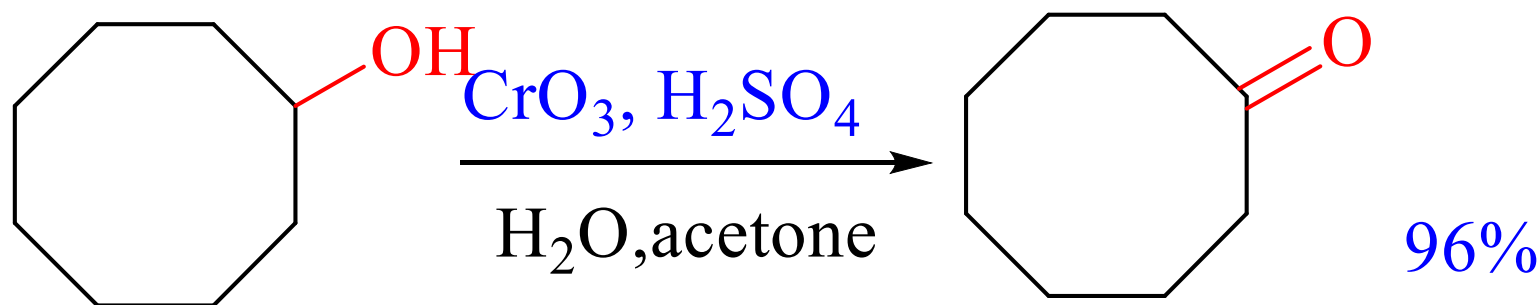
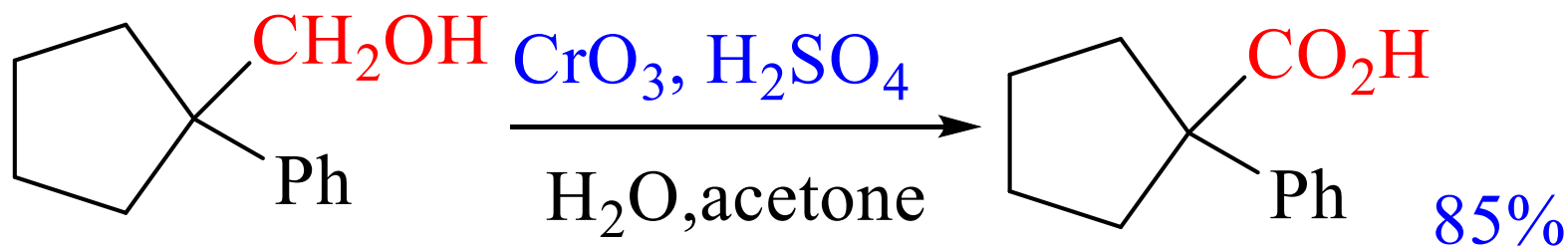
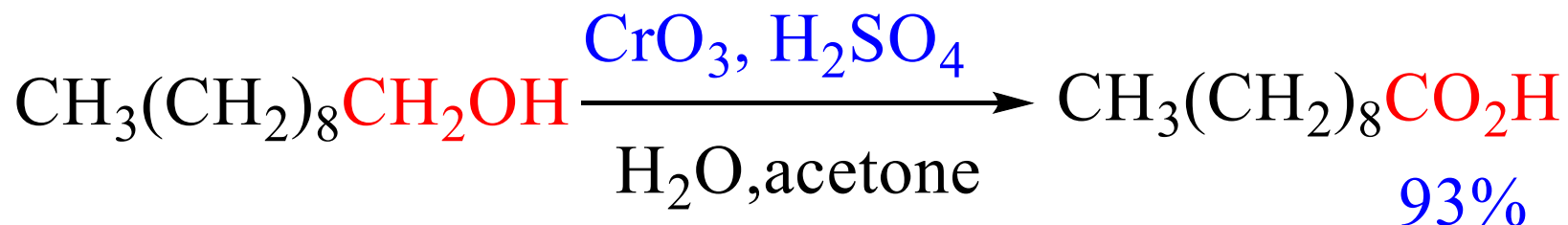


氧化剂: KMnO_4 , H_2CrO_4 , $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$
 $\text{Na}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$, $\text{CrO}_3/\text{H}_2\text{SO}_4$

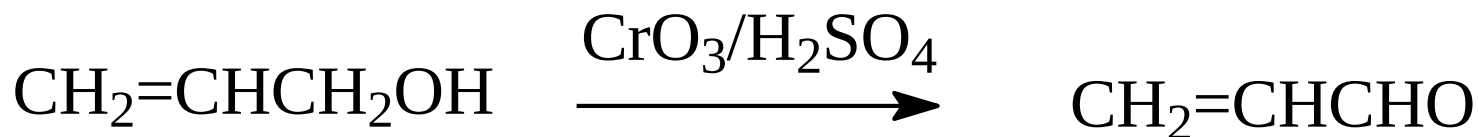
Example



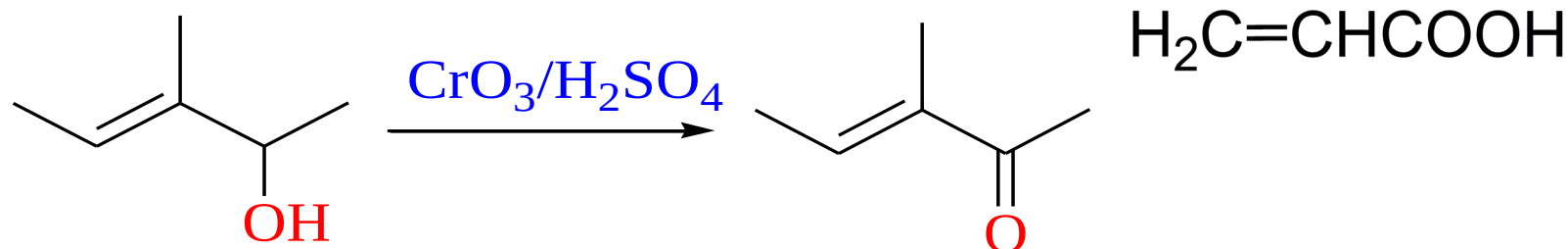
Jones' reagent — $\text{CrO}_3 / \text{H}_2\text{SO}_4$



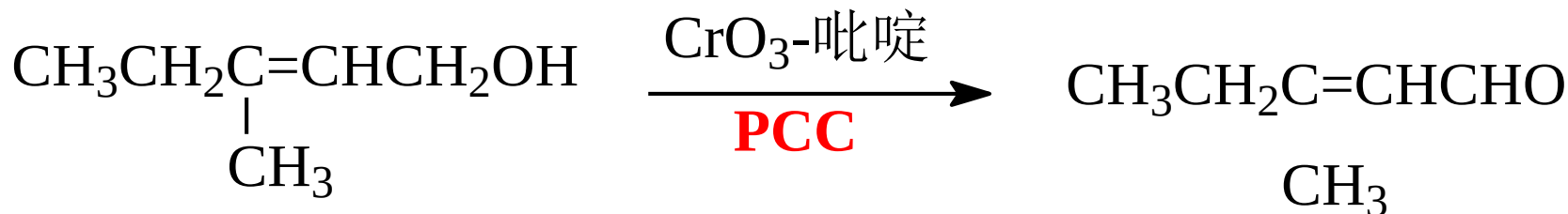
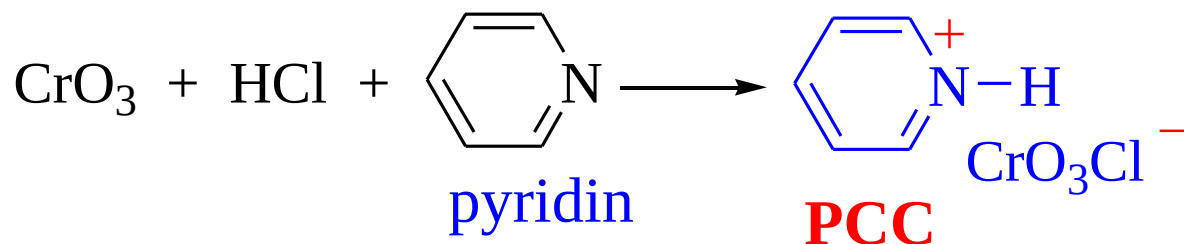
Johns试剂: $\text{CrO}_3/\text{H}_2\text{SO}_4$



丙烯醛沸点较低，可以蒸馏得到；如不蒸馏，得到丙烯酸。

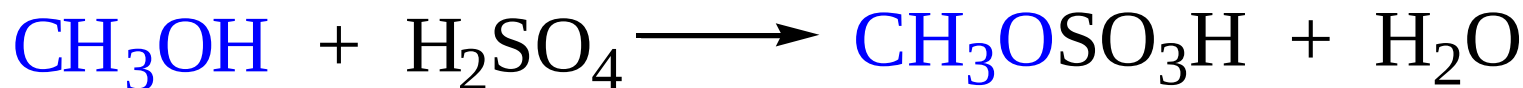
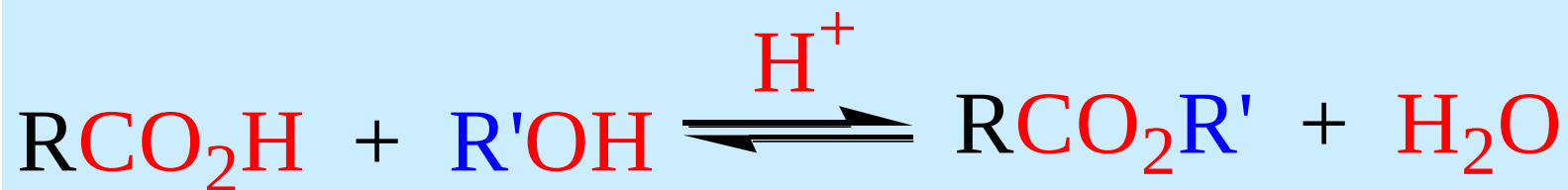


PCC (Pyridinium Chlorochromate) 氯铬酸吡啶盐

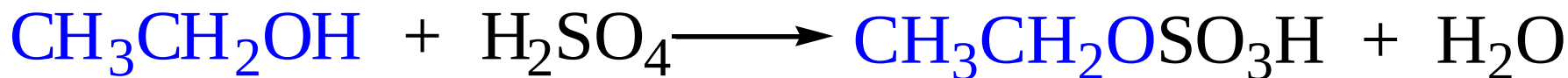


成酯反应

醇与含氧酸(有机酸、无机酸)反应成酯

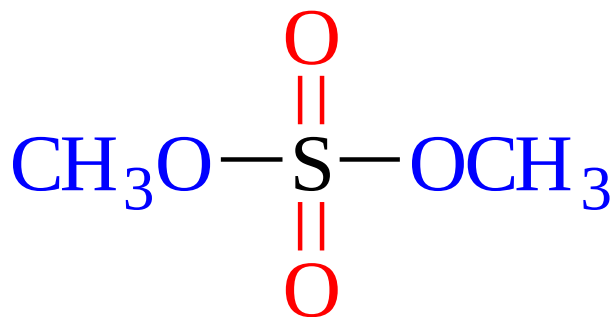
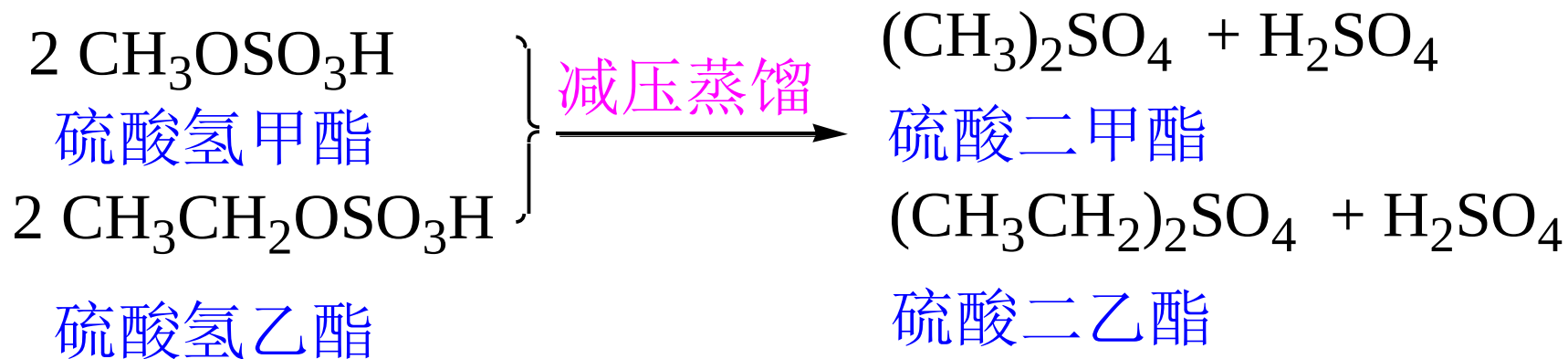


硫酸氢甲酯



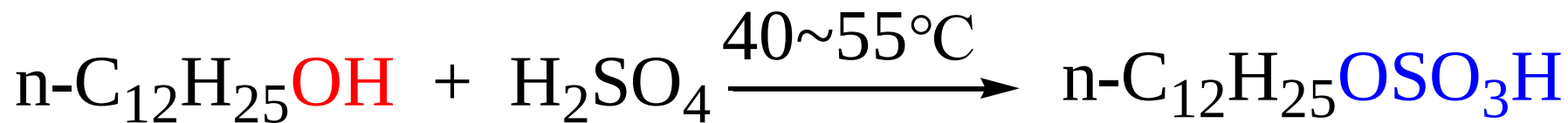
硫酸氢乙酯

用途



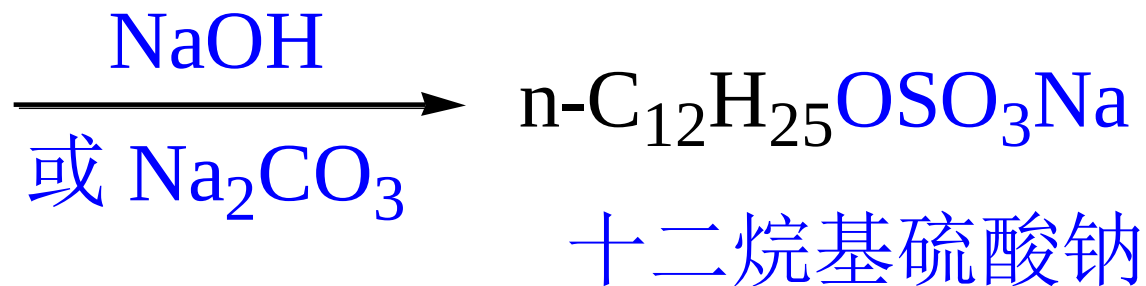
剧毒!

用途



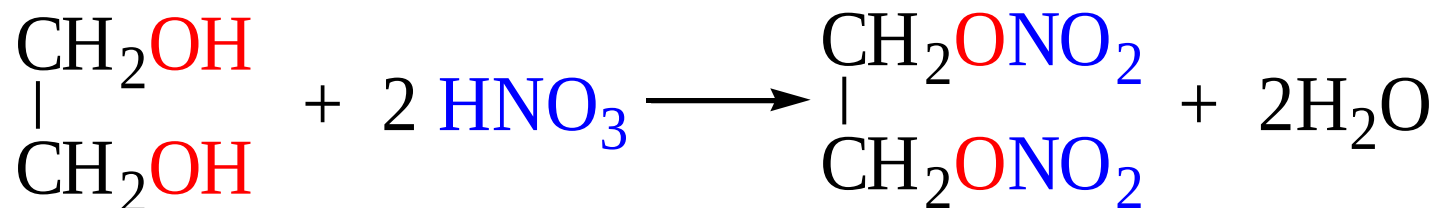
月桂醇

十二烷基硫酸



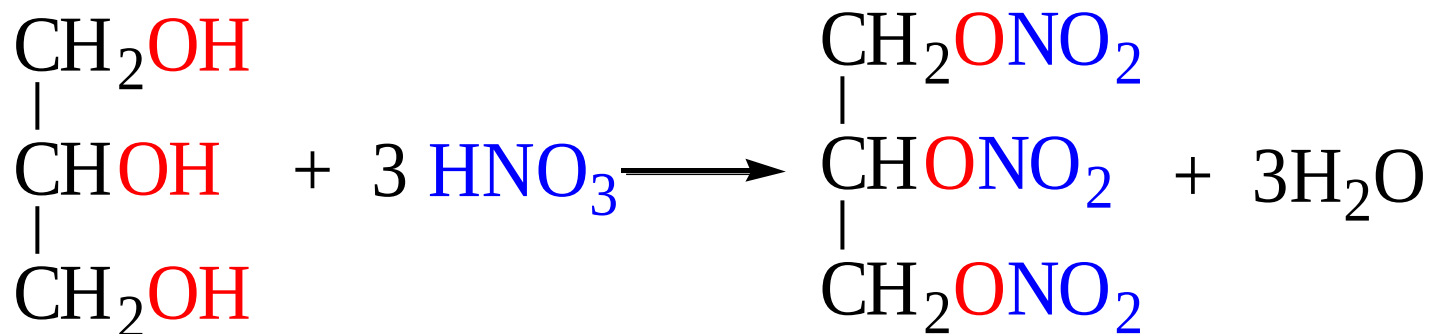
乳化剂

用途



乙二醇

乙二醇二硝酸酯



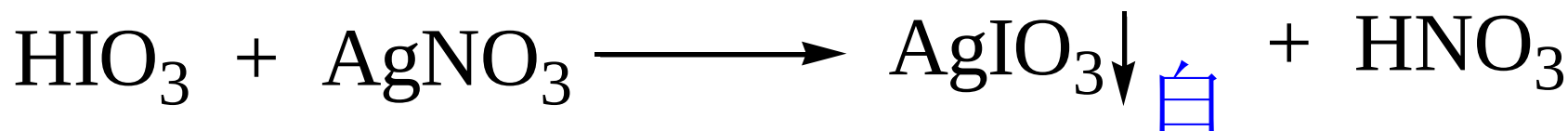
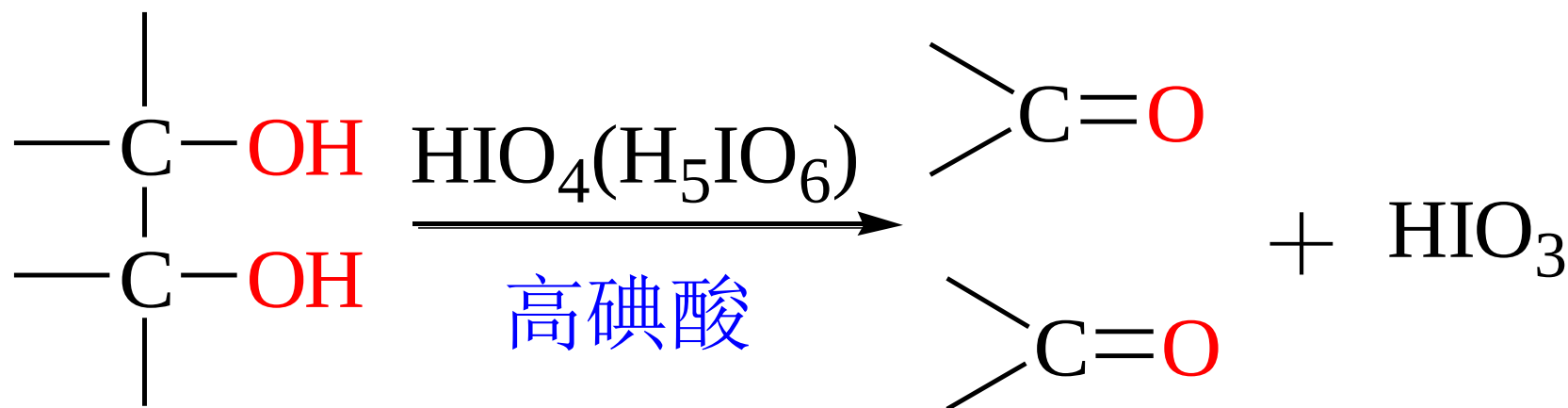
甘油

甘油三硝酸酯
硝化甘油

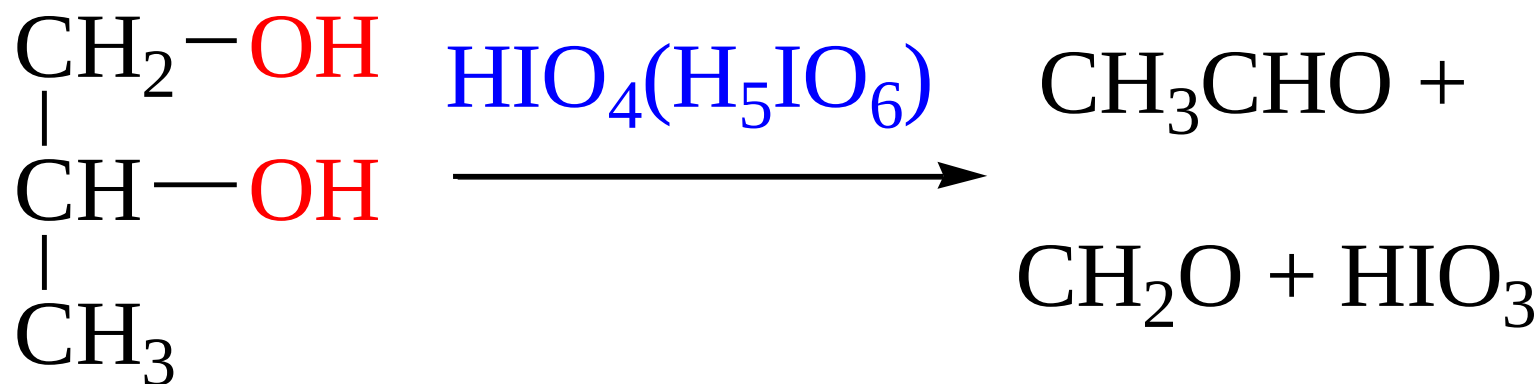
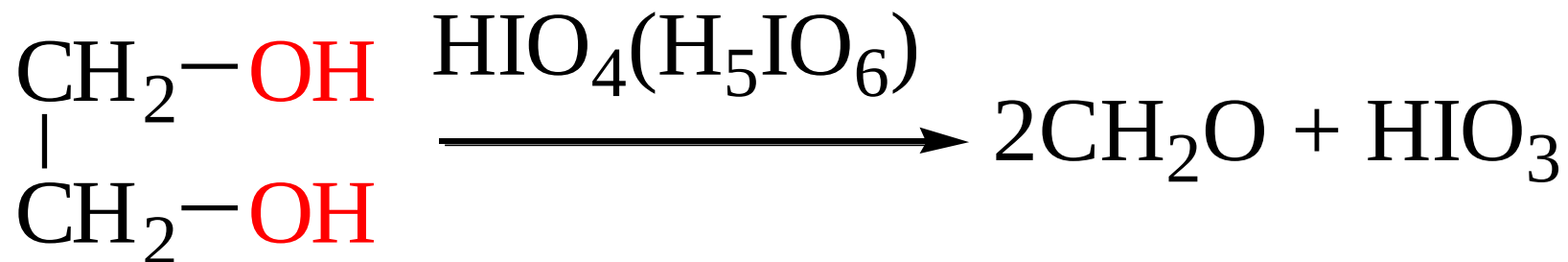
烈性炸药

邻二醇的反应

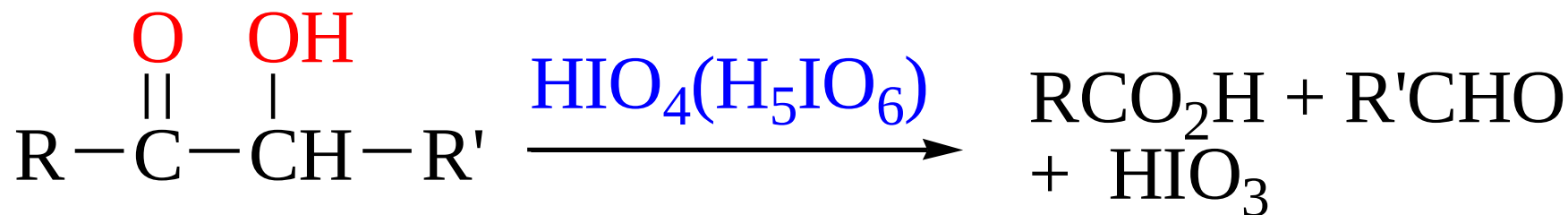
1. 被高碘酸氧化



Example

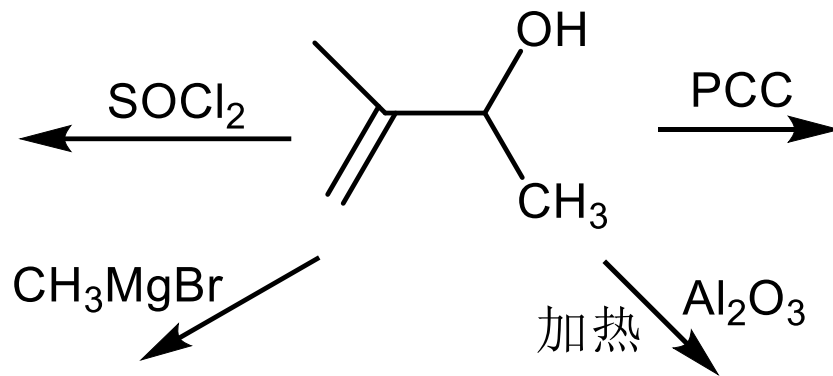


Example

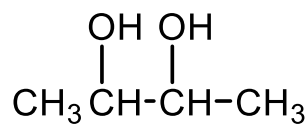


测试

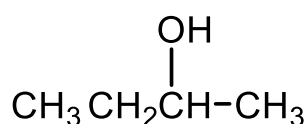
完成反应



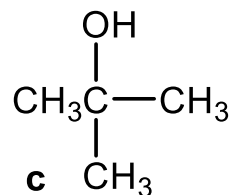
用化学方法鉴别如下化合物



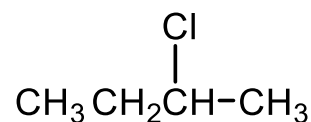
a



b



c

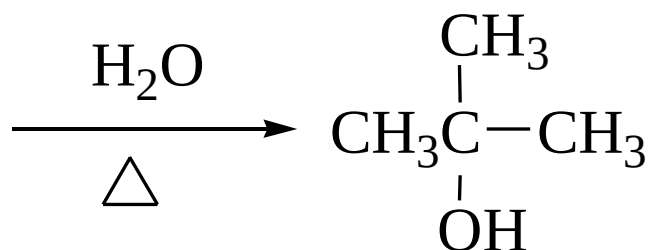
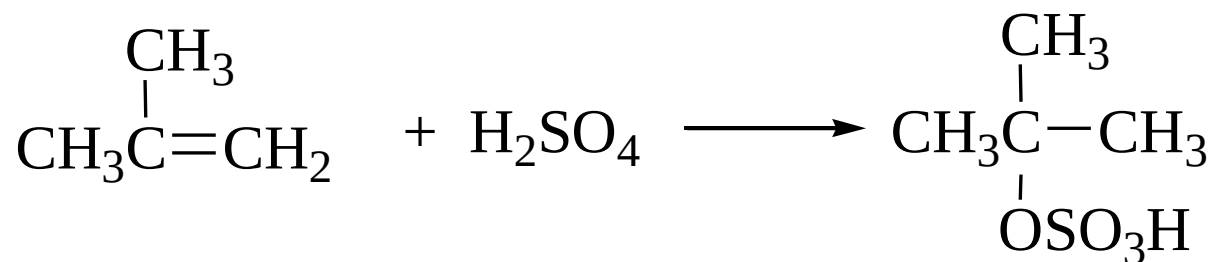
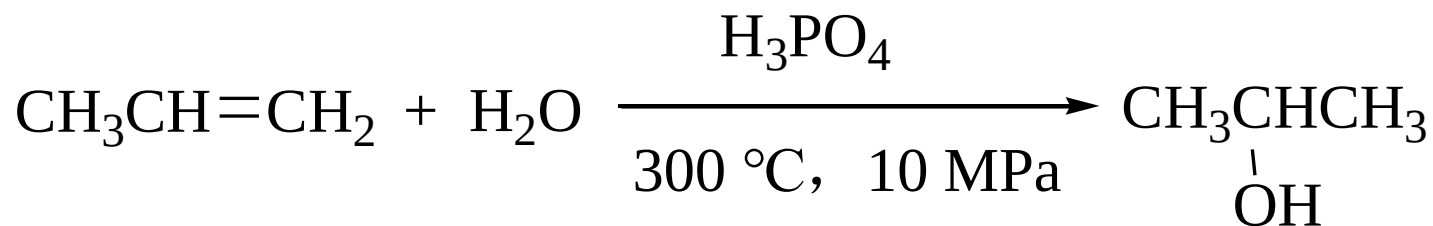


d

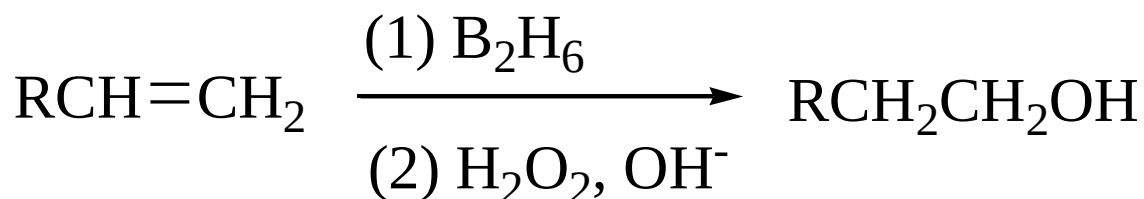
醇的制备和来源

- 由烯烃制备——水合，硼氢化-氧化
- 由羰基化合物制备——Grignard 反应，与炔化物反应，还原
- 由卤代烃水解

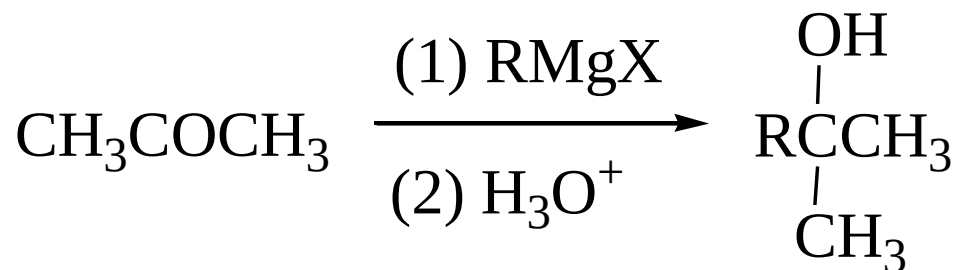
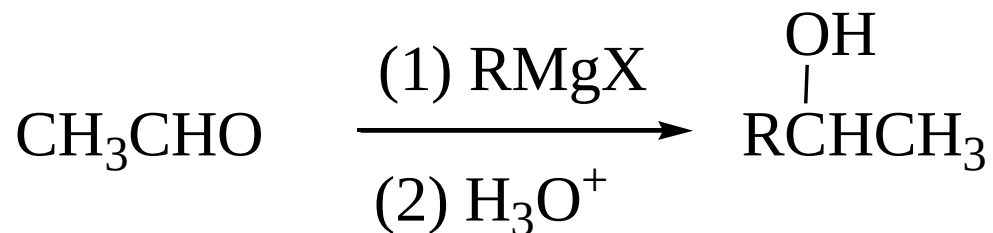
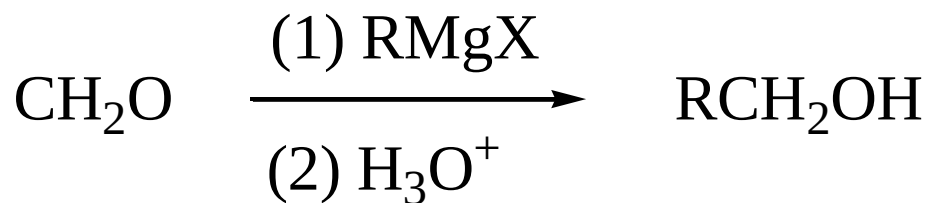
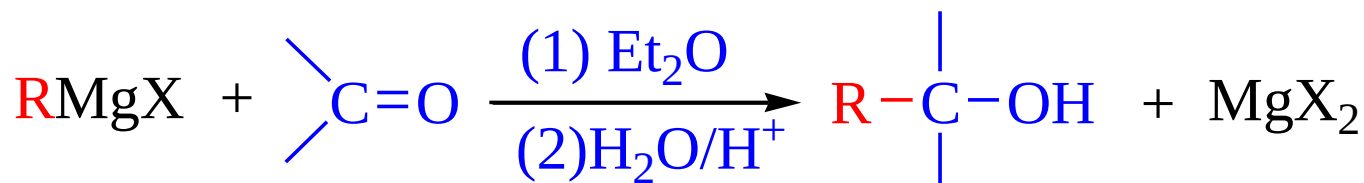
由烯烃制备



可能发生重排

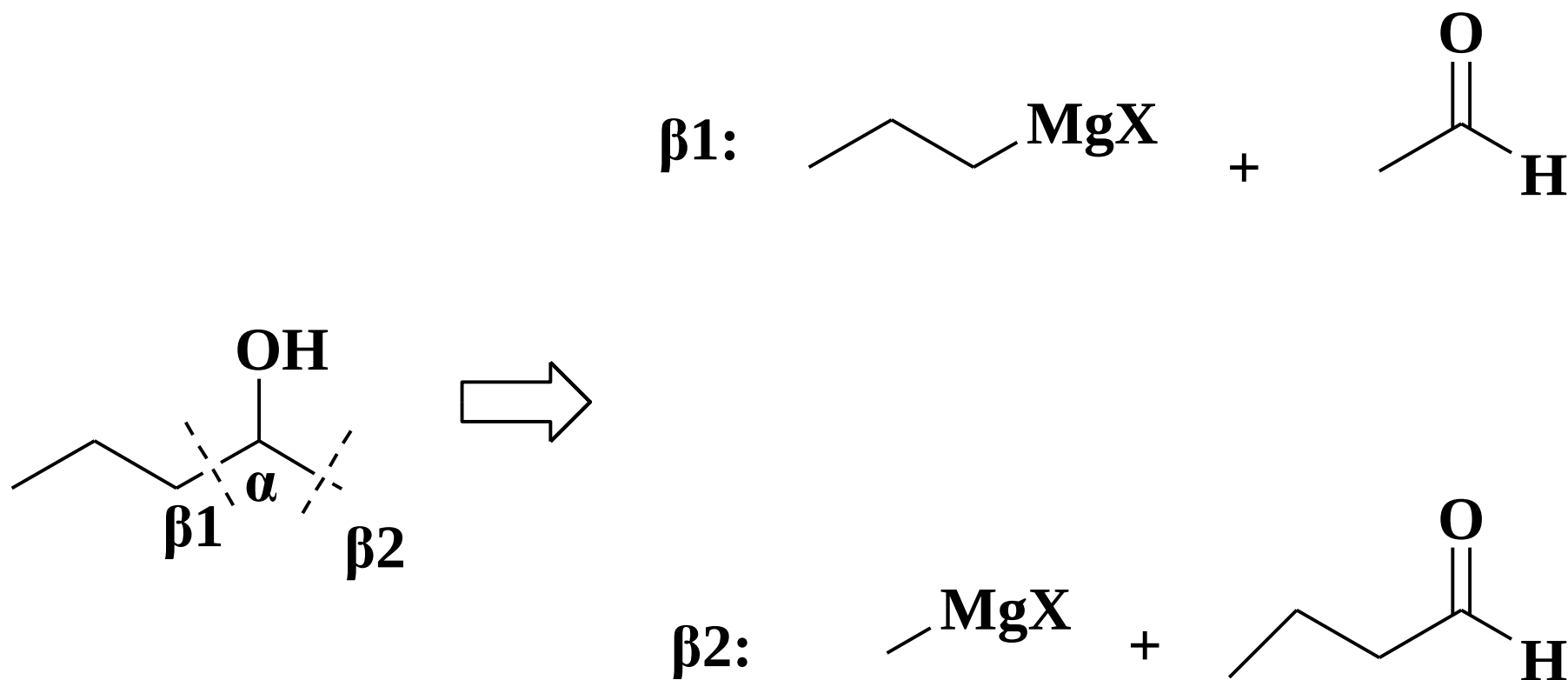


由羰基化合物制备

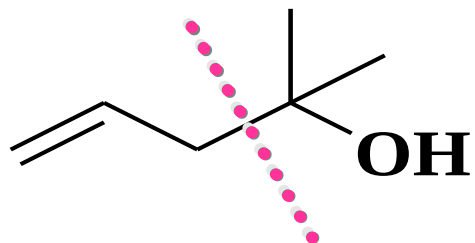


Example

逆向合成分析



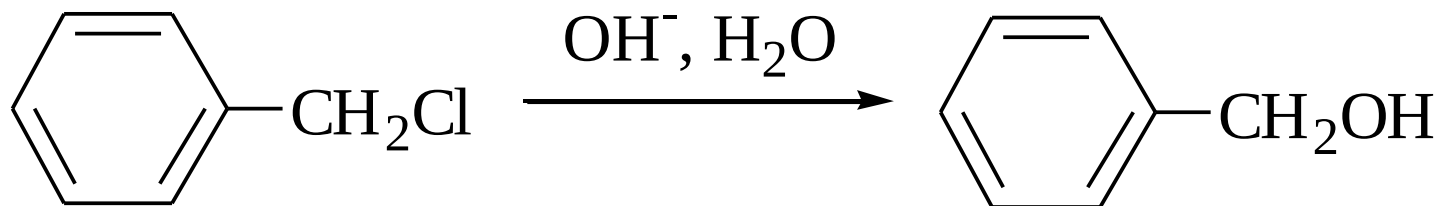
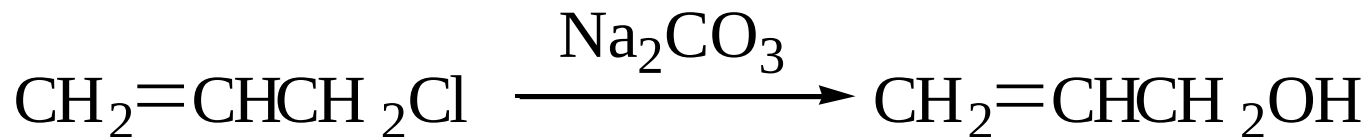
Example



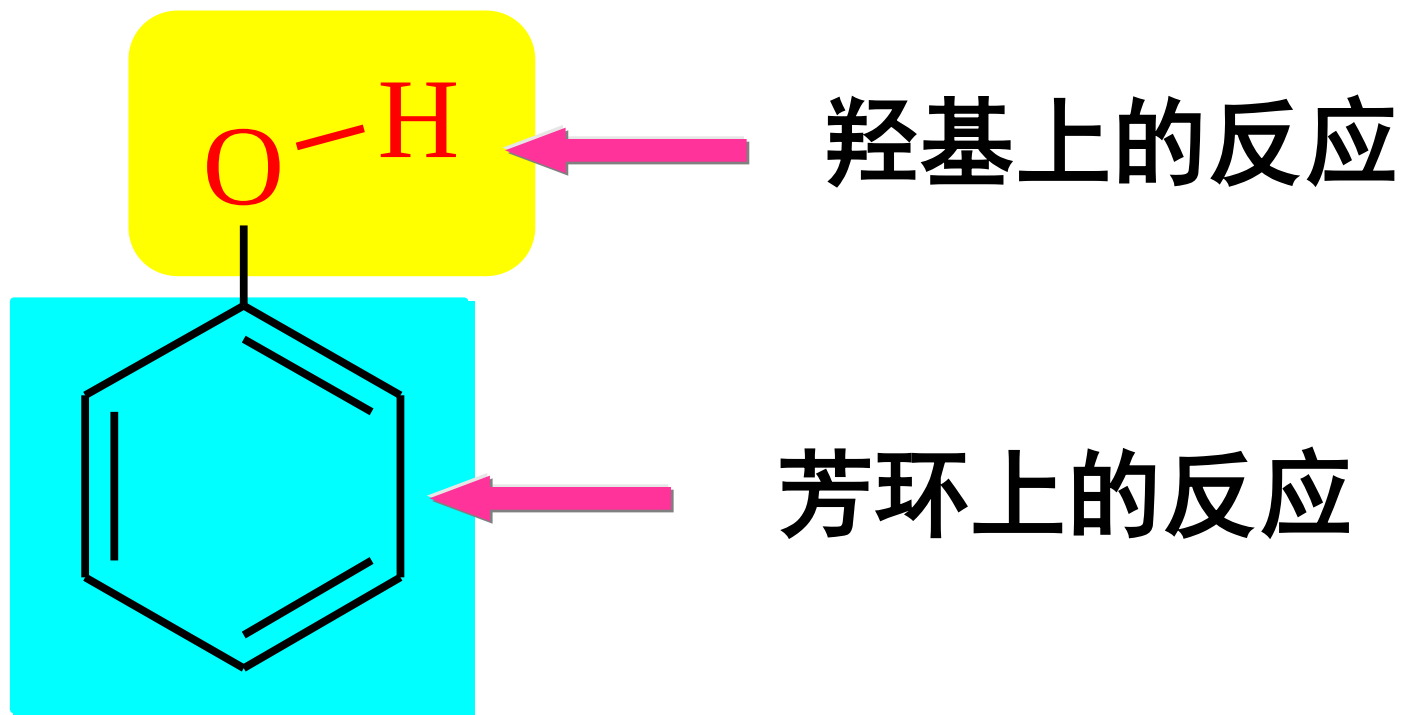
由卤代烃水解

- 副反应多
- 一般情况下醇比卤代烃易得

由卤代烃水解主要应用到如下类型的化合物中：

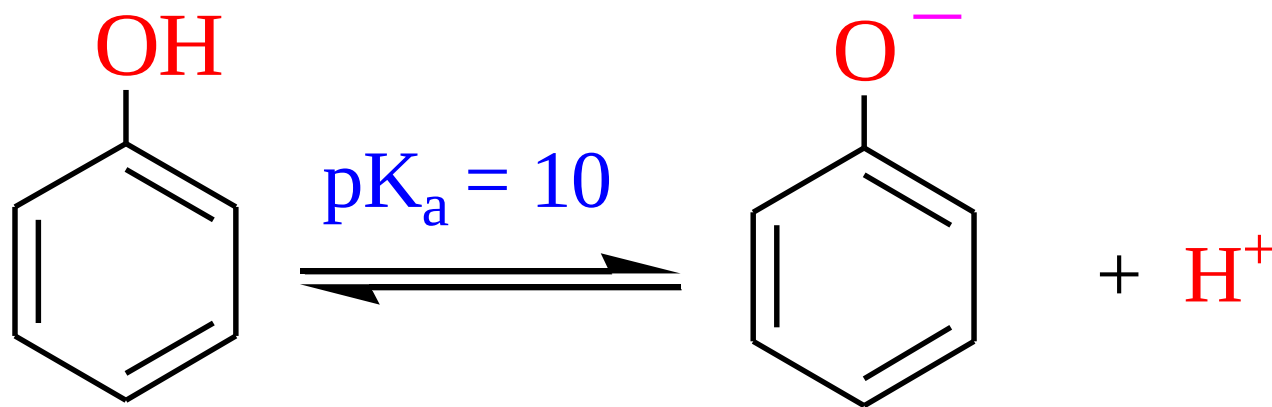


酚的化学性质

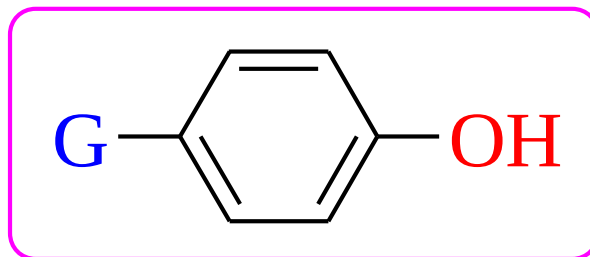


羟基上的反应

1. 酚的酸性



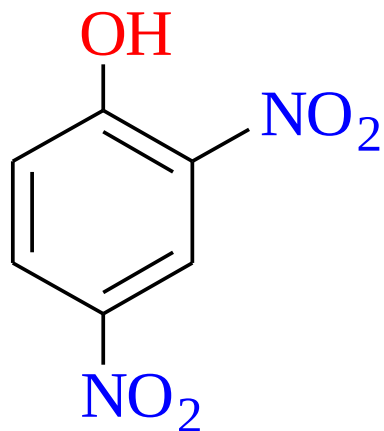
取代酚的酸性



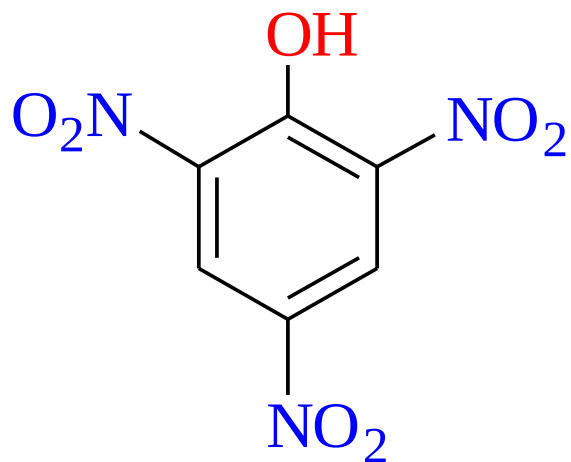
G	NH ₂	CH ₃ O	CH ₃	H	Cl	Br	I	NO ₂
pK _a	10.46	10.21	10.17	10.0	9.38	9.35	9.30	7.15



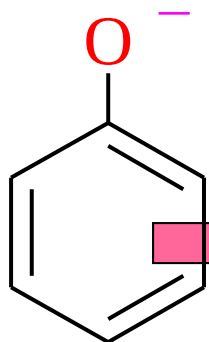
	NO ₂	<i>m</i> -	<i>o</i> -	<i>p</i> -
	pK _a	8.28	7.17	7.15



pK_a 4.09



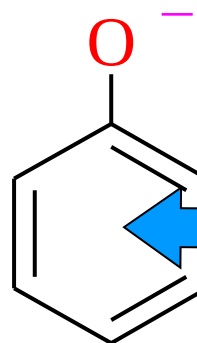
0.25



吸电子取代基

负离子稳定性提高

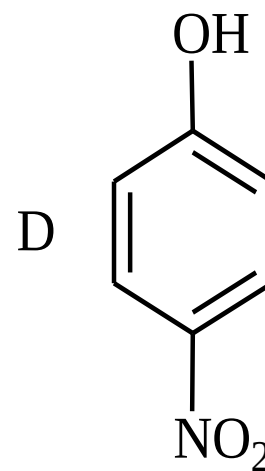
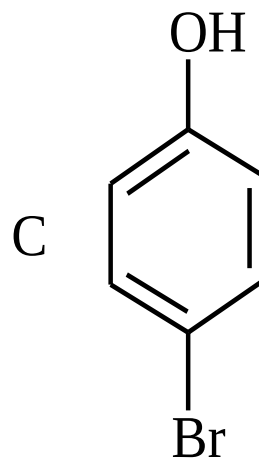
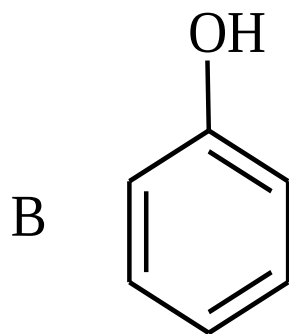
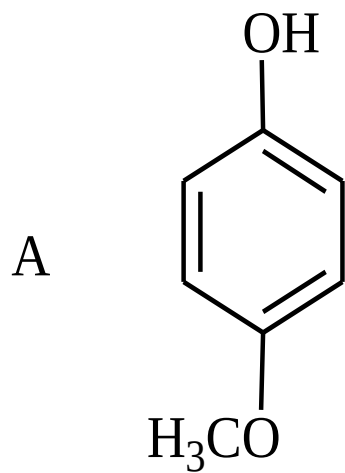
酸性增强



给电子取代基

负离子稳定性降低

酸性减弱

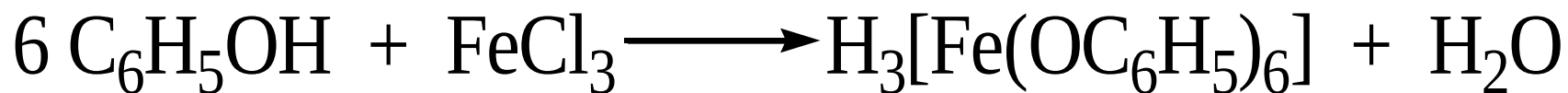


酸性比较

羟基上的反应

2. 酚与三氯化铁的显色反应

六苯酚合铁酸

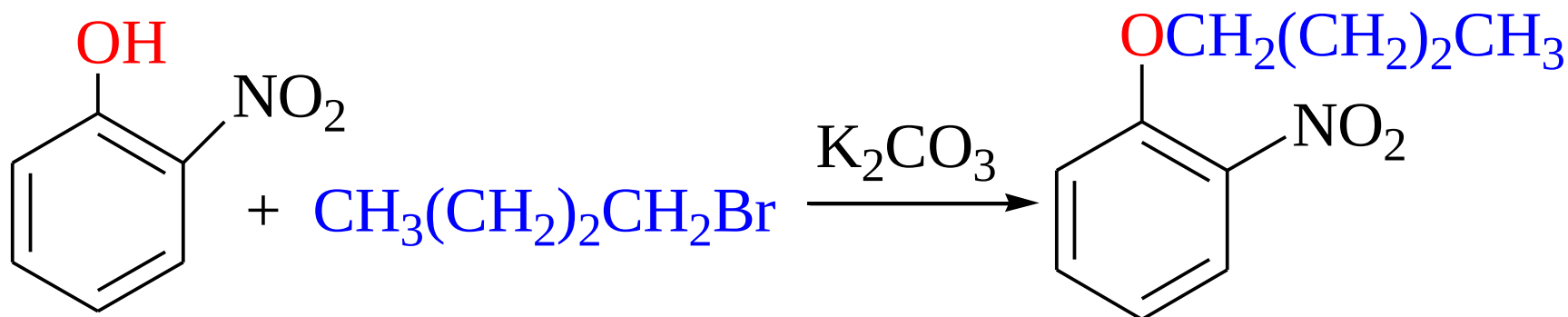
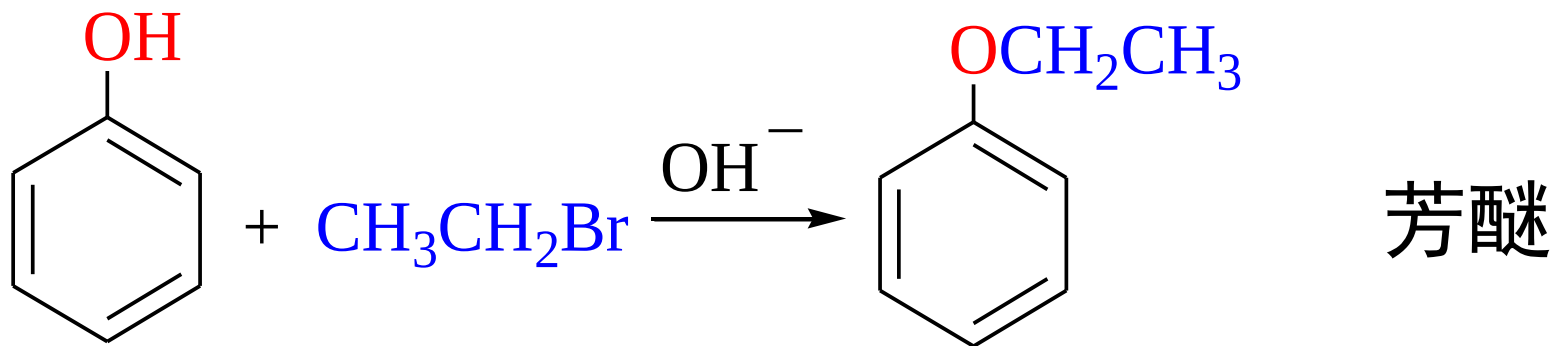


蓝紫色

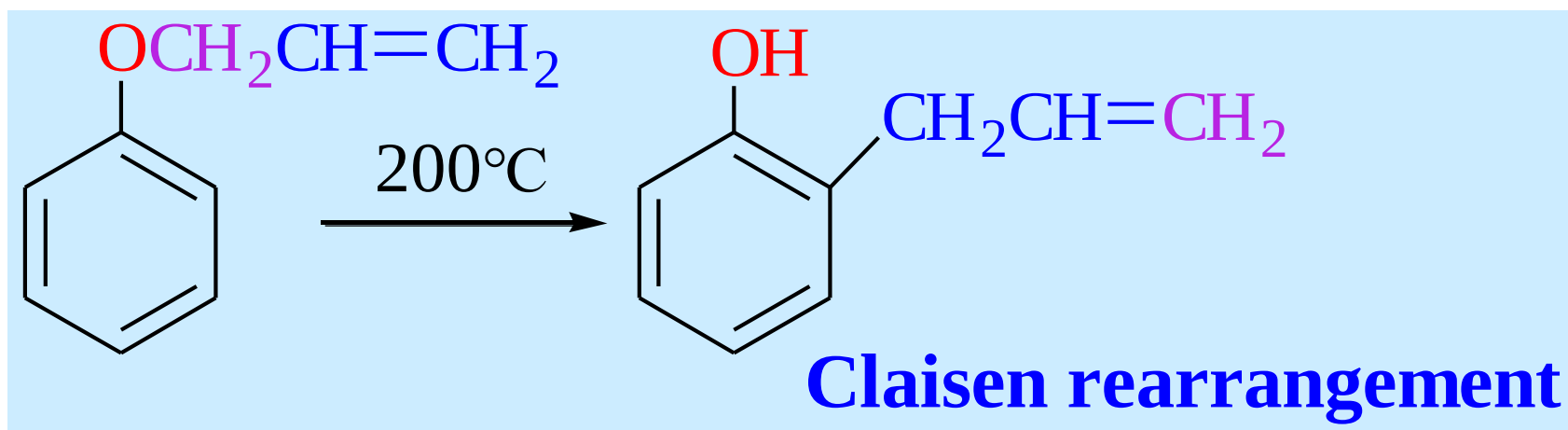
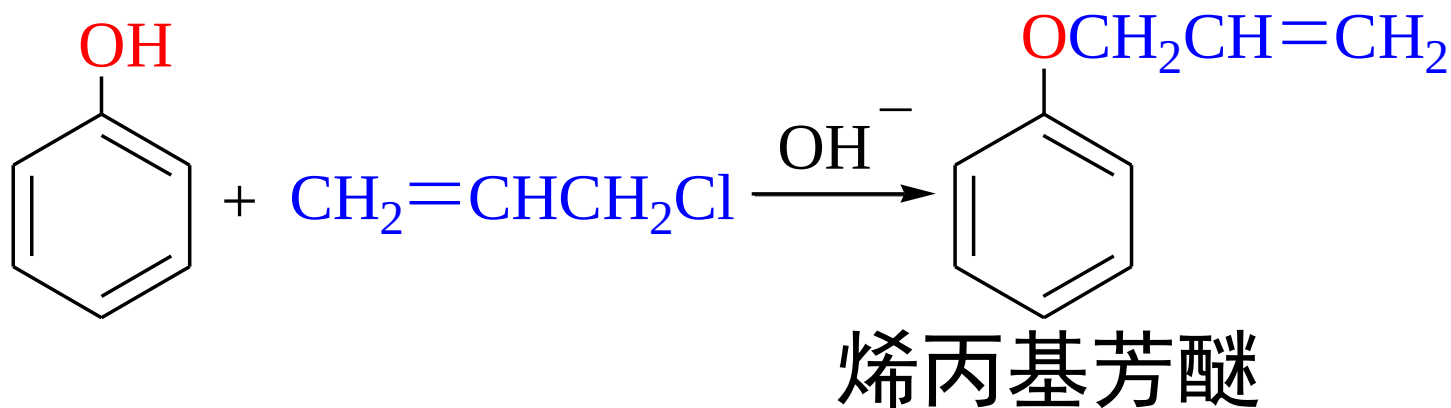
烯醇式结构的化合物大多能使三氯化铁显色

羟基上的反应

3. 烷基化反应和Claisen重排

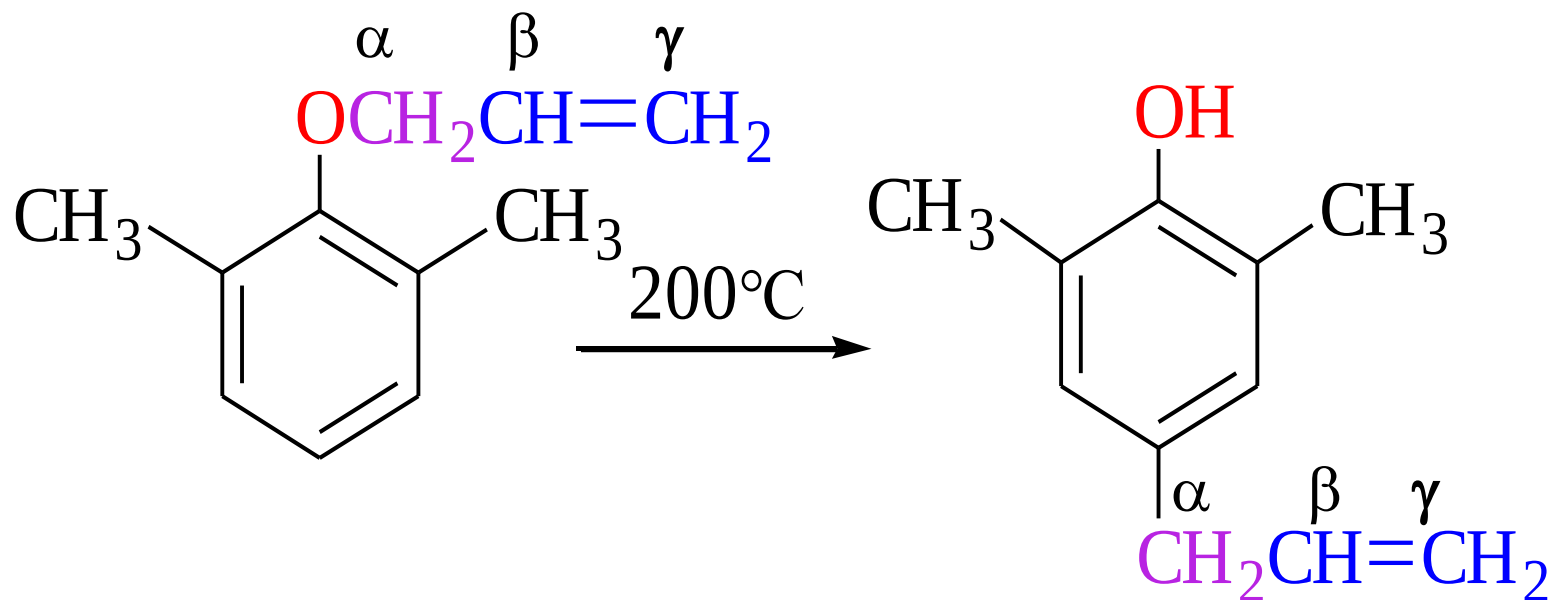
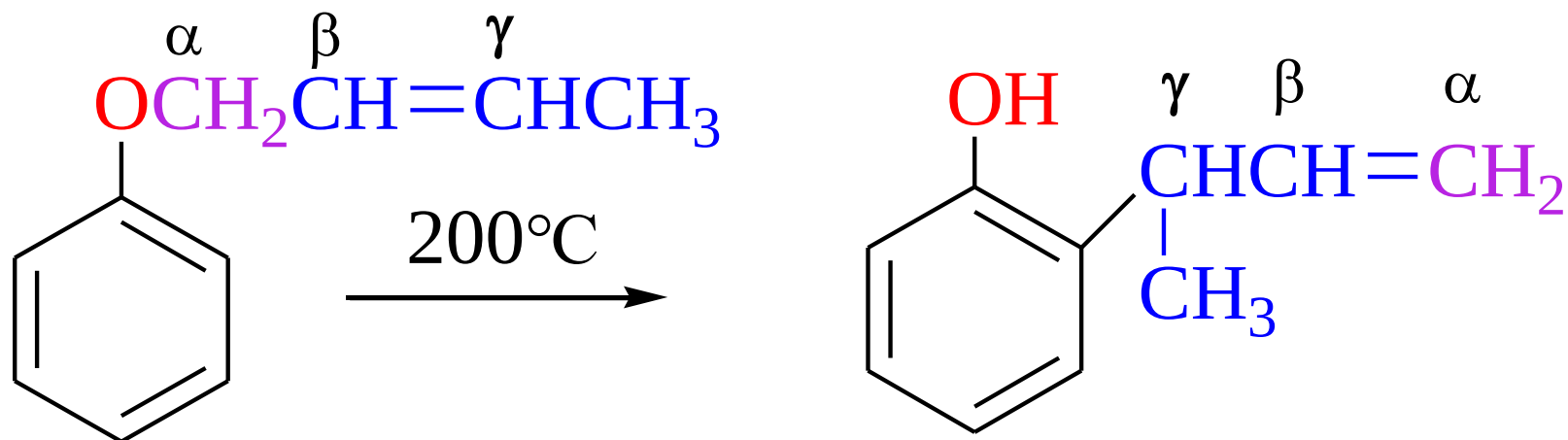


Claisen重排

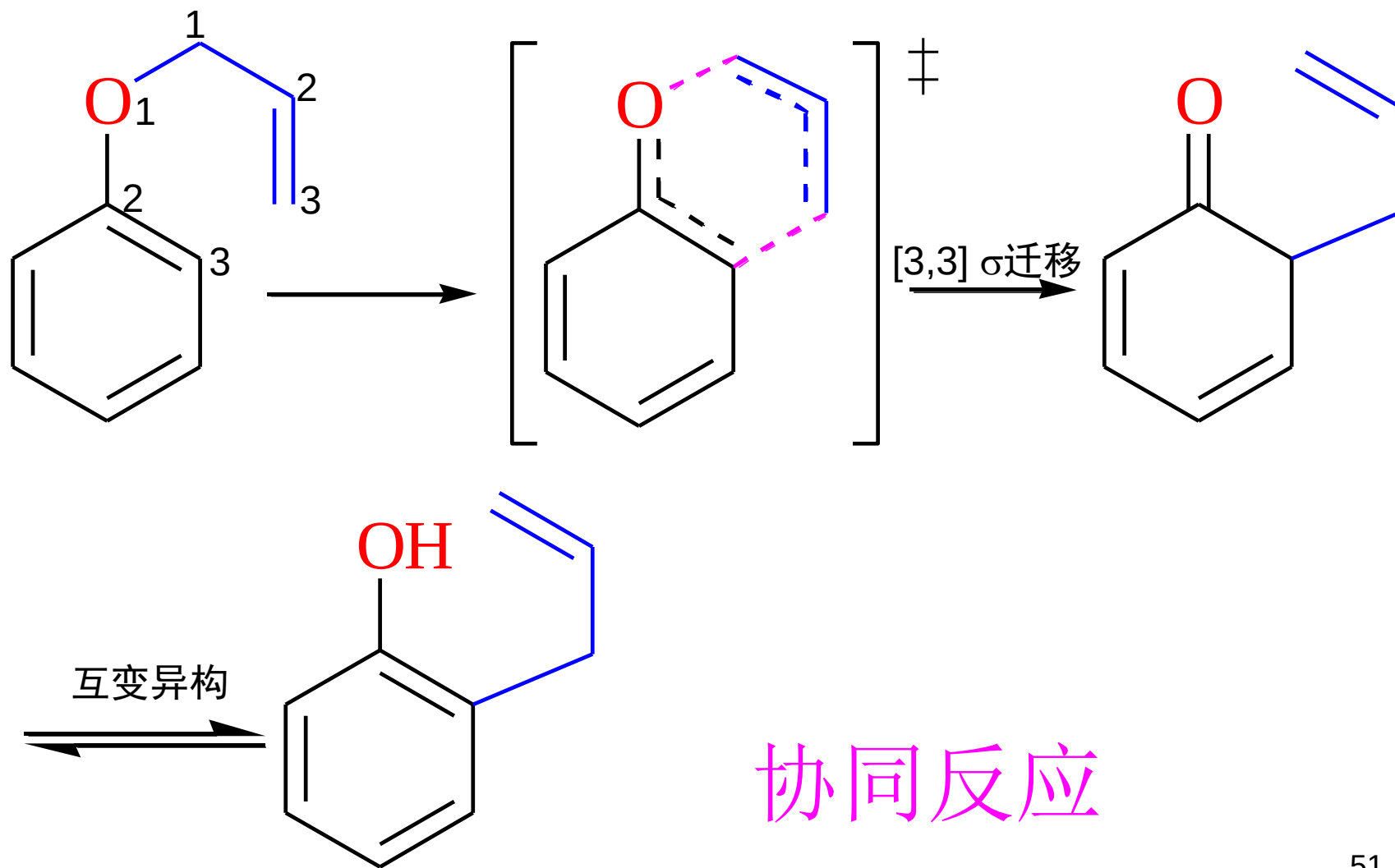


克莱森重排

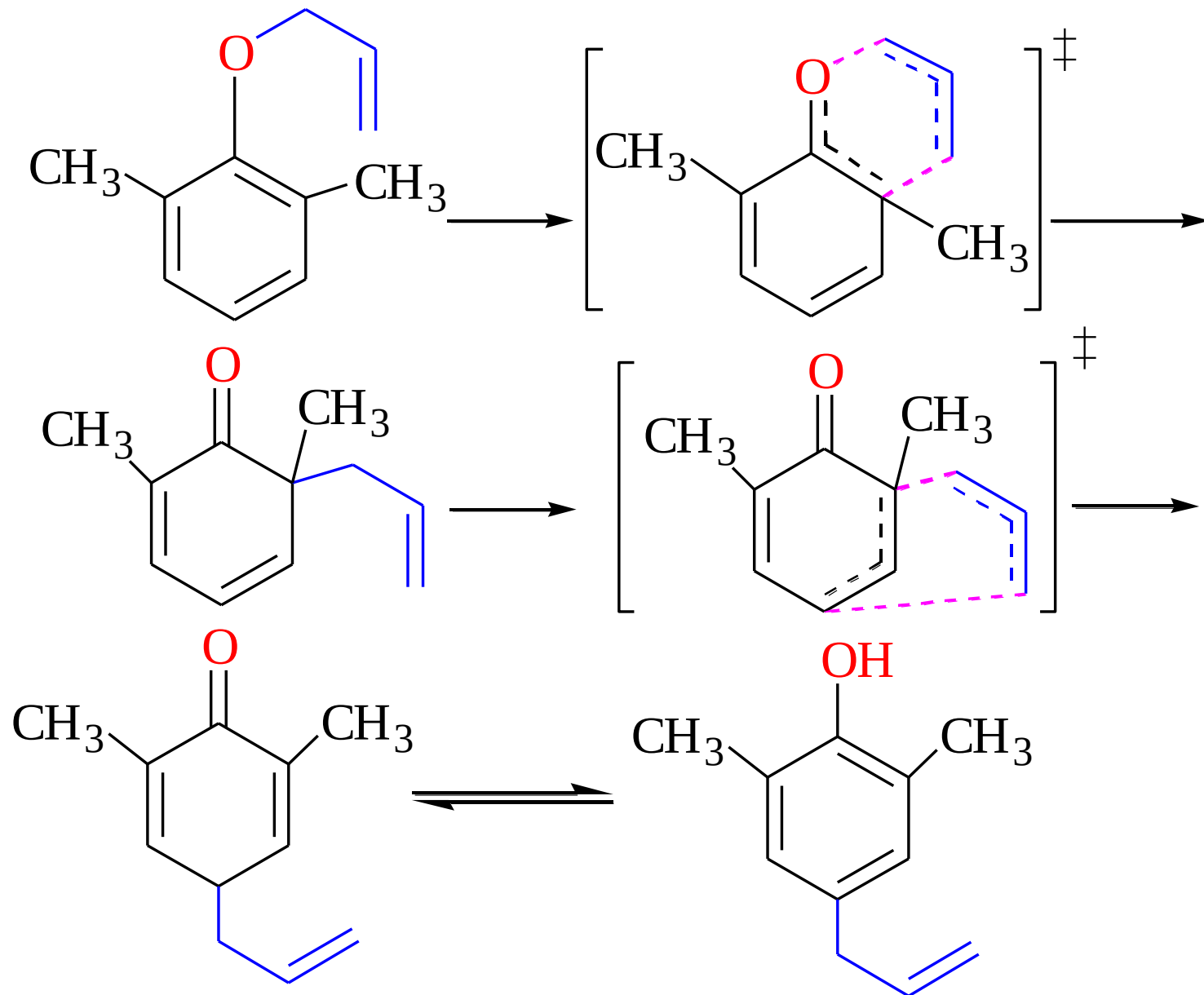
Example



反应机理



反应机理

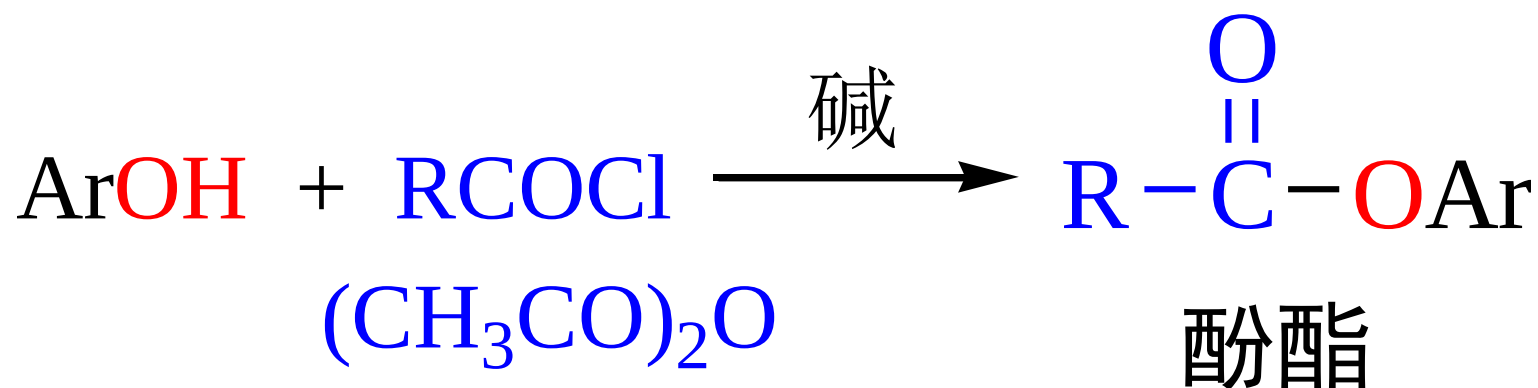


羟基上的反应

弗赖斯

4. 酰基化反应及Fries重排

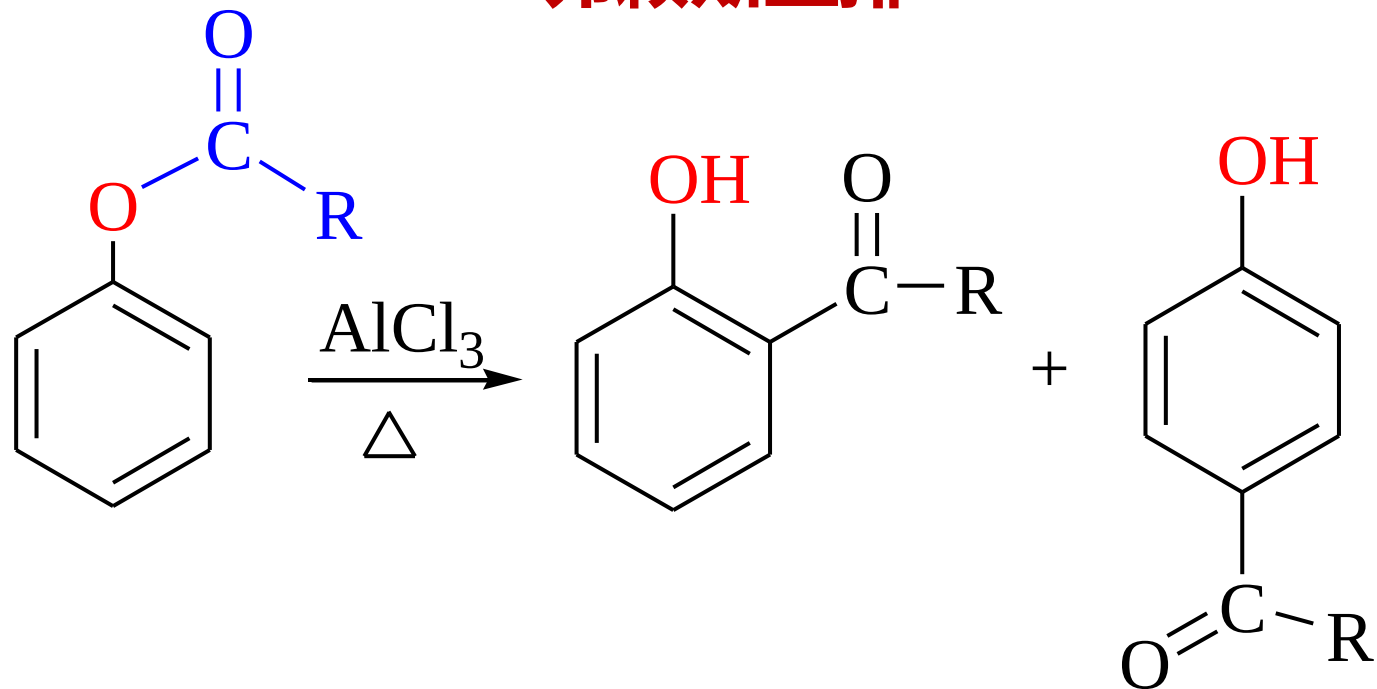
一般用酰氯或酸酐反应



碱：吡啶，NaOH，Na₂CO₃，NaOAc

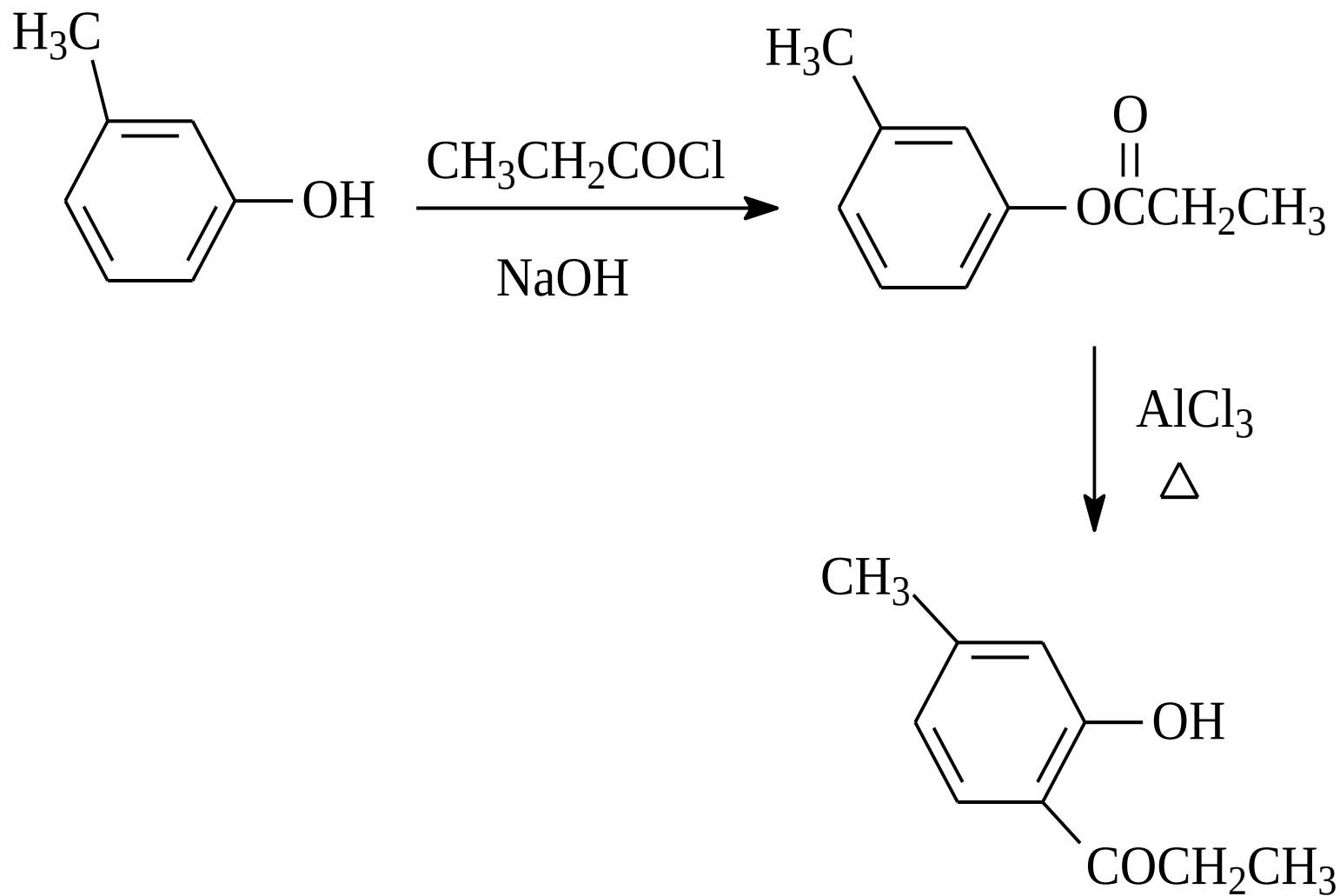
Fries重排

弗赖斯重排



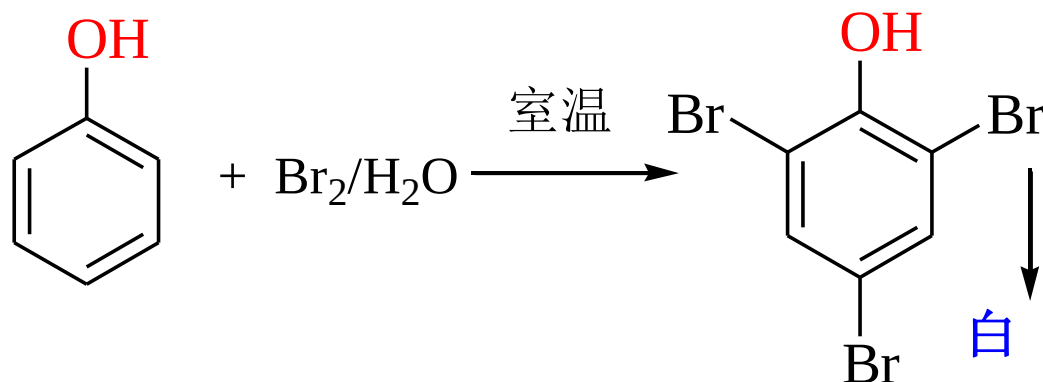
高温下产物以邻位为主
较低温度下产物以对位为主

Example

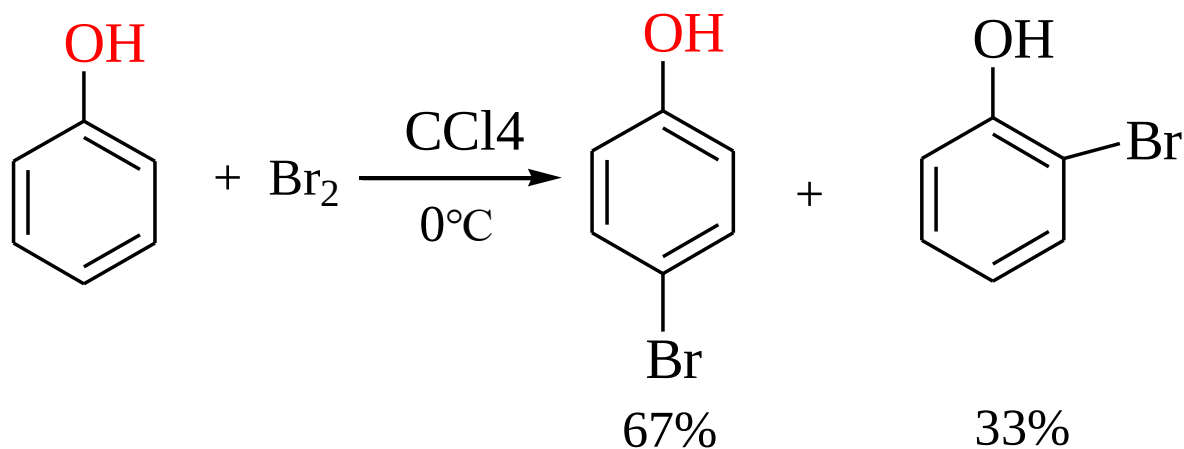


芳环上的取代反应(自学)

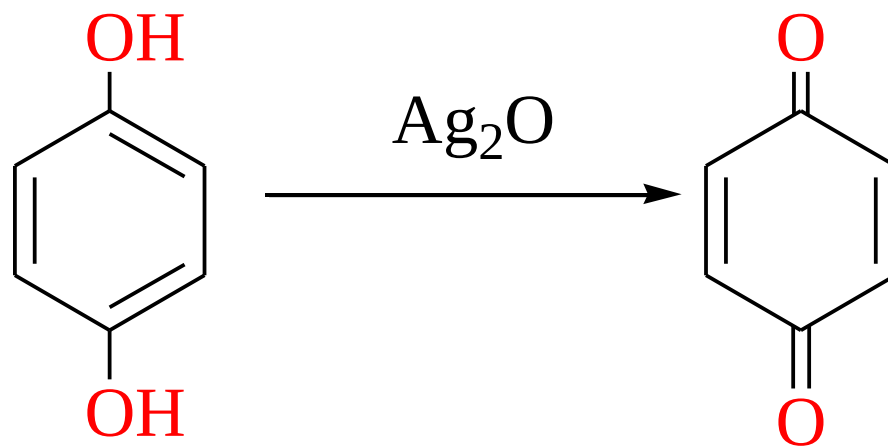
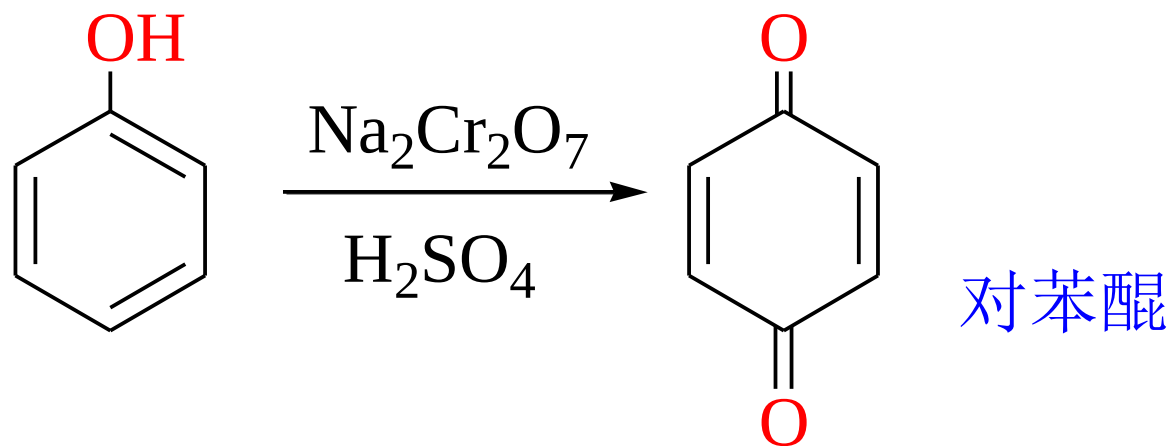
卤化反应



可用于鉴别苯酚

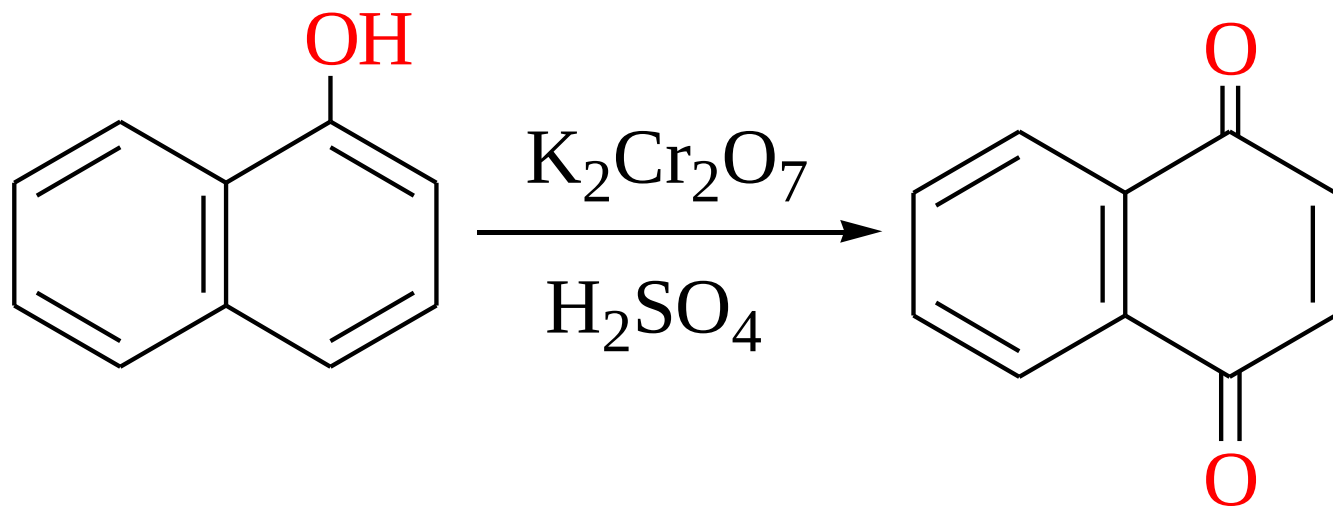


酚的氧化反应

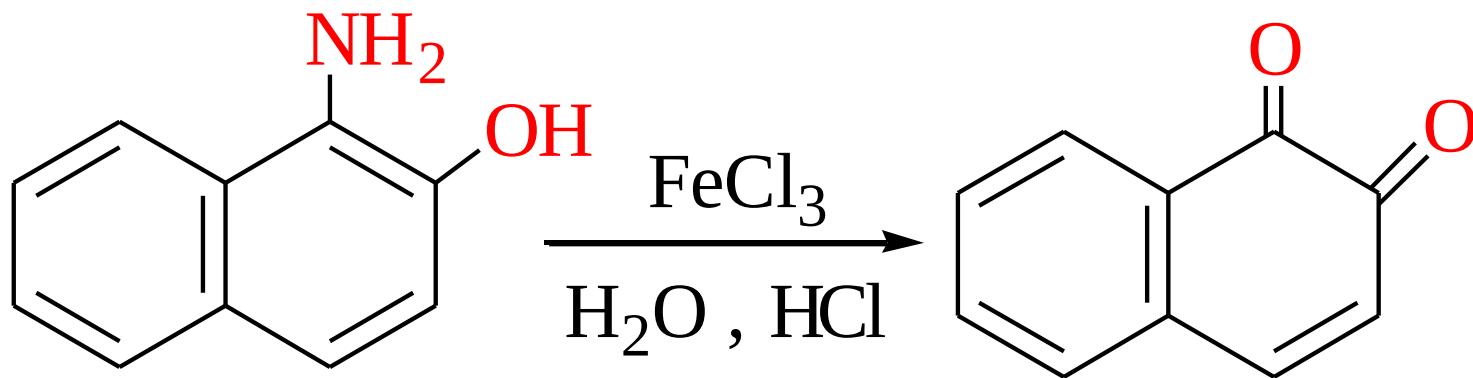


对苯二酚

Example

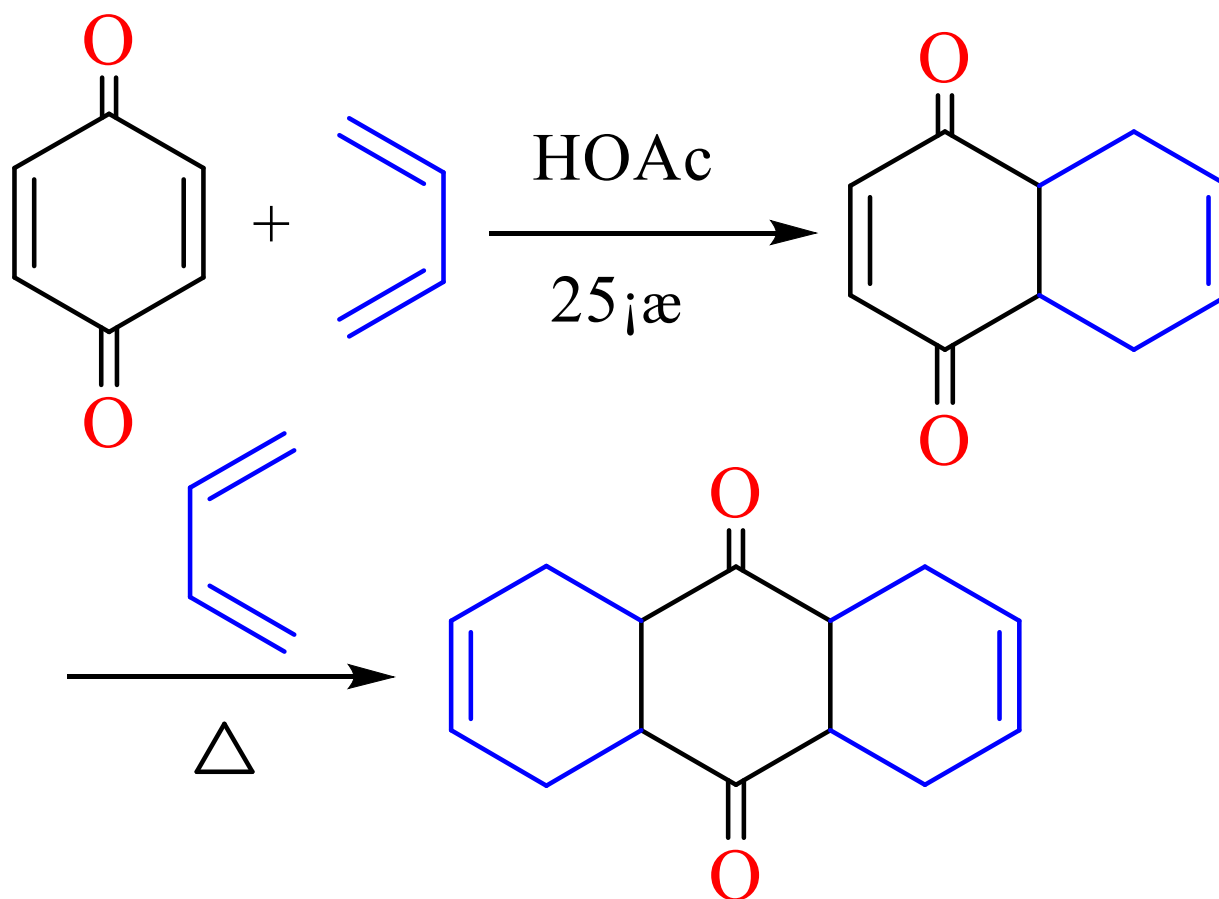


1,4-萘醌



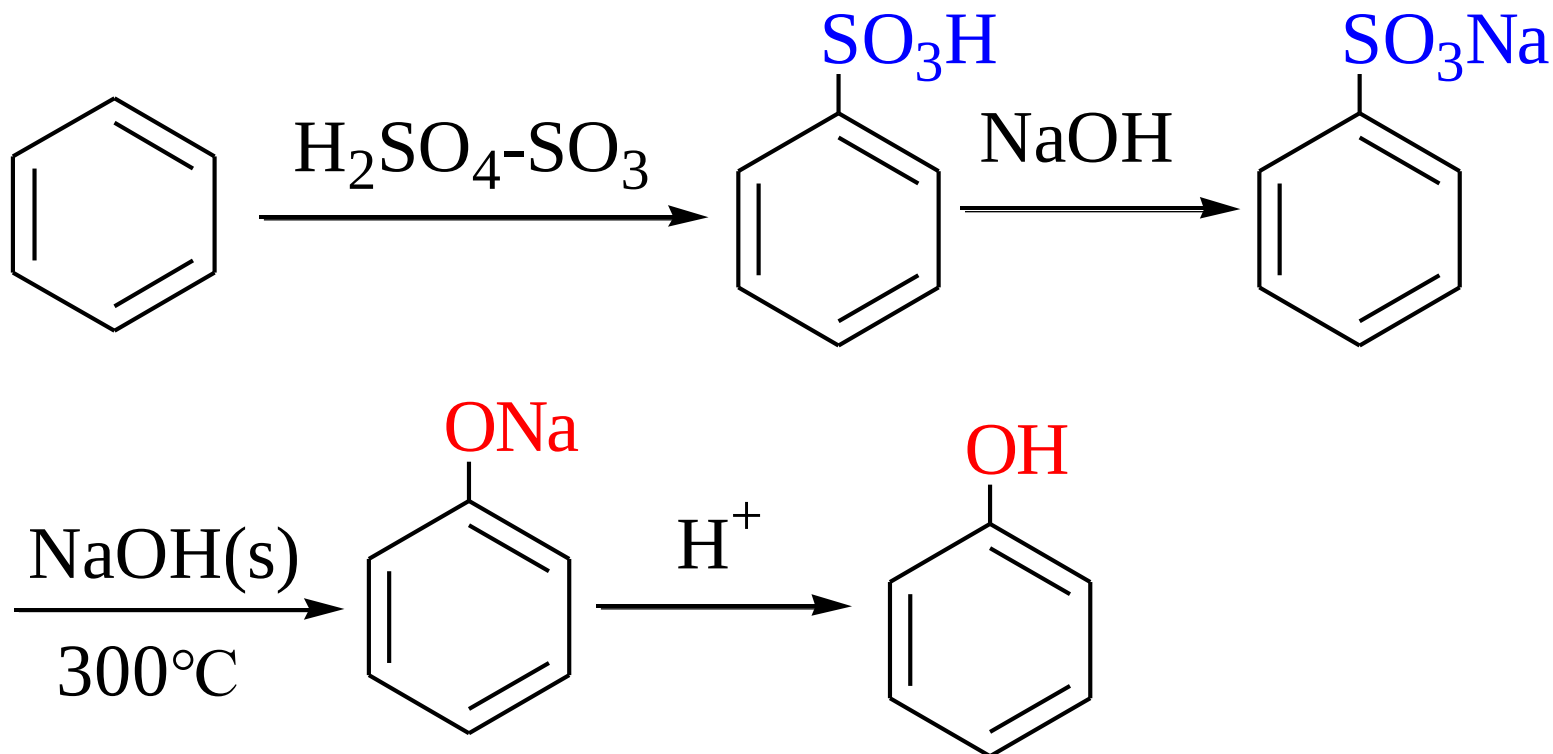
1,2-萘醌

对苯醌可作为亲双烯体 发生Diels—Alder反应

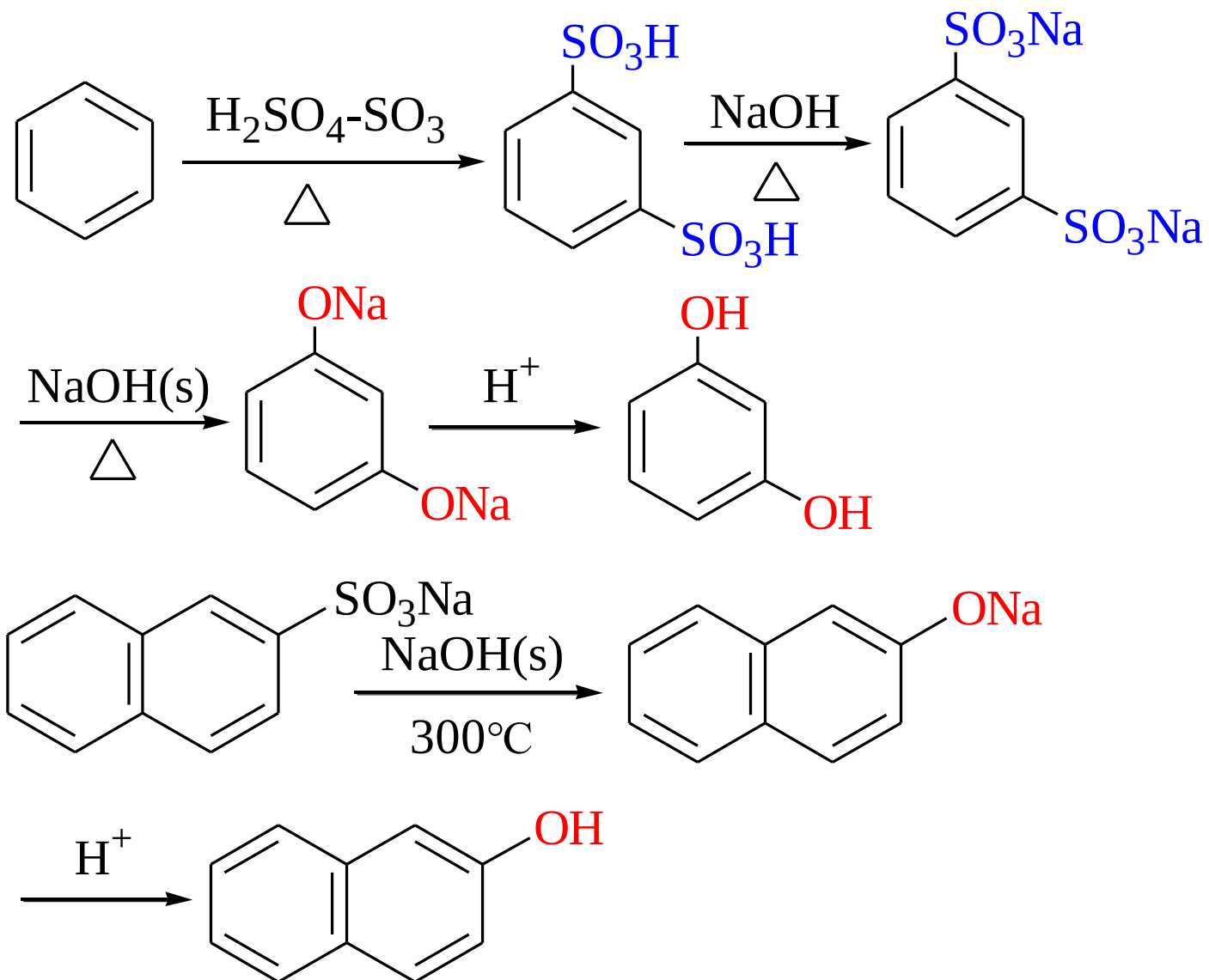


酚的制备

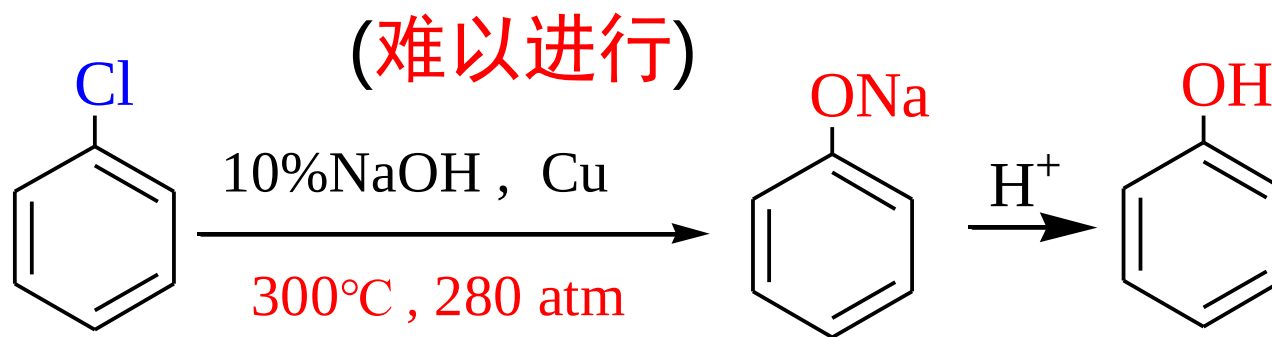
1. 磺酸盐碱熔法



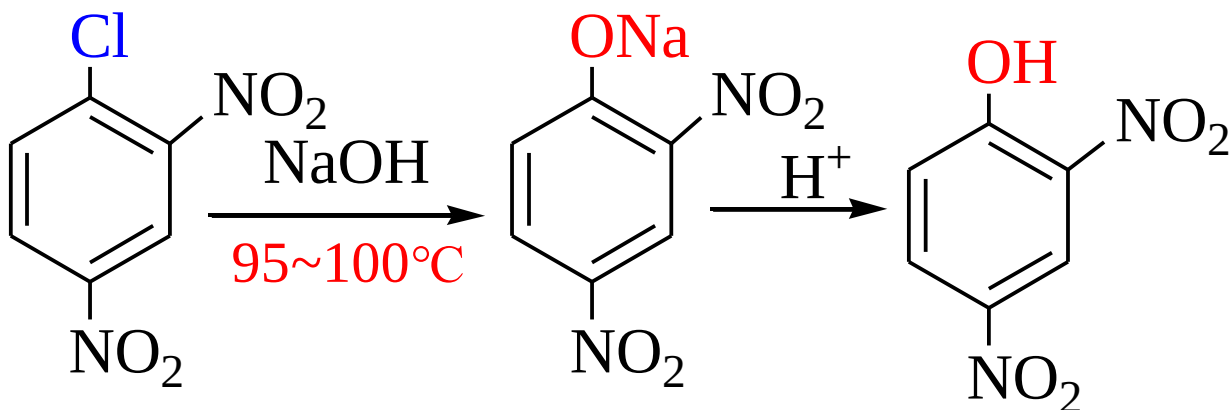
Example



2. 氯苯水解法

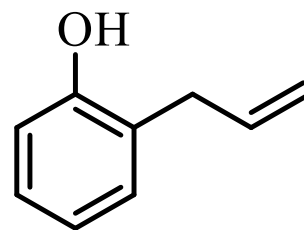
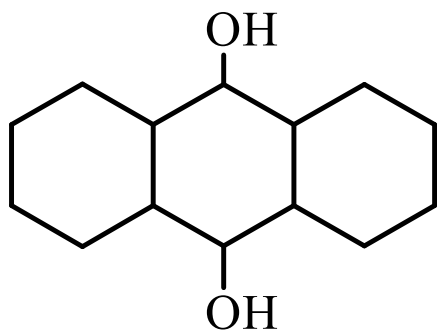


卤素的邻、对位有强吸电子基团时，
水解变得容易进行：



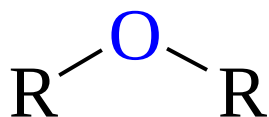
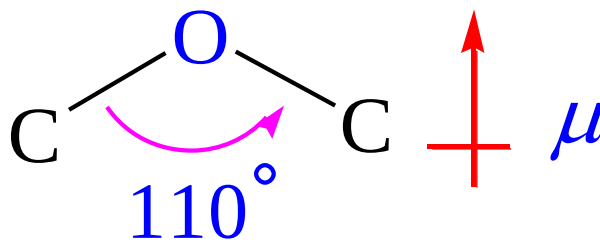
测试

以苯酚为原料，合成下列物质

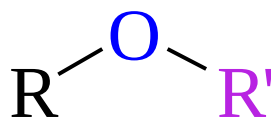


醚

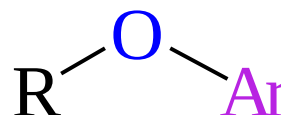
醚键



简单醚
(对称醚)

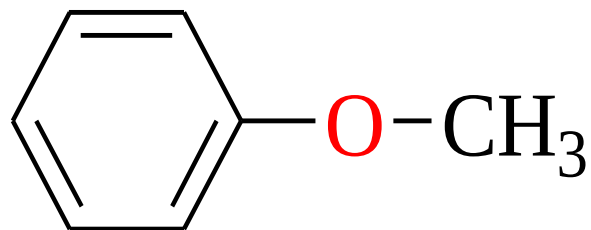


混合醚
(不对称醚)

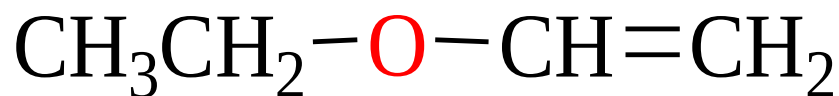


芳醚

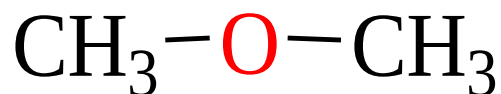
命名（了解）



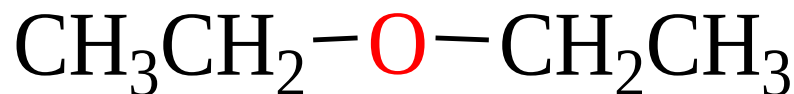
苯甲醚(茴香醚)



乙基乙烯基醚



二甲醚(甲醚)



二乙醚(乙醚)

命名（了解）

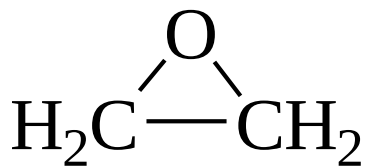
2. 环醚

环醚

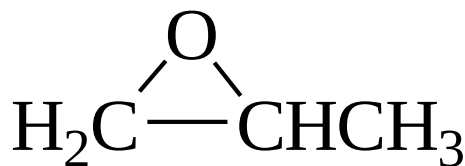
环氧化合物

大环多醚

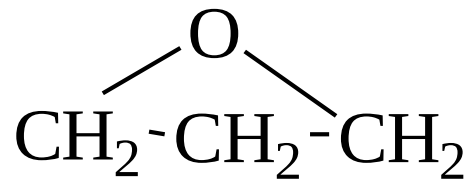
环氧化合物



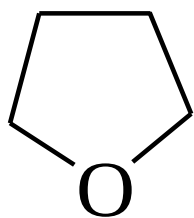
环氧乙烷



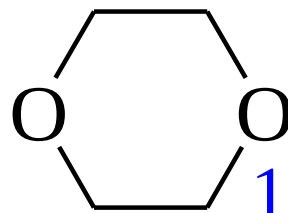
环氧丙烷



1,3-环氧丙烷

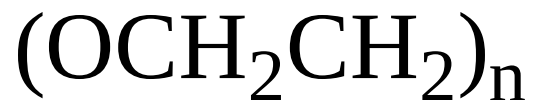


四氢呋喃
(THF)

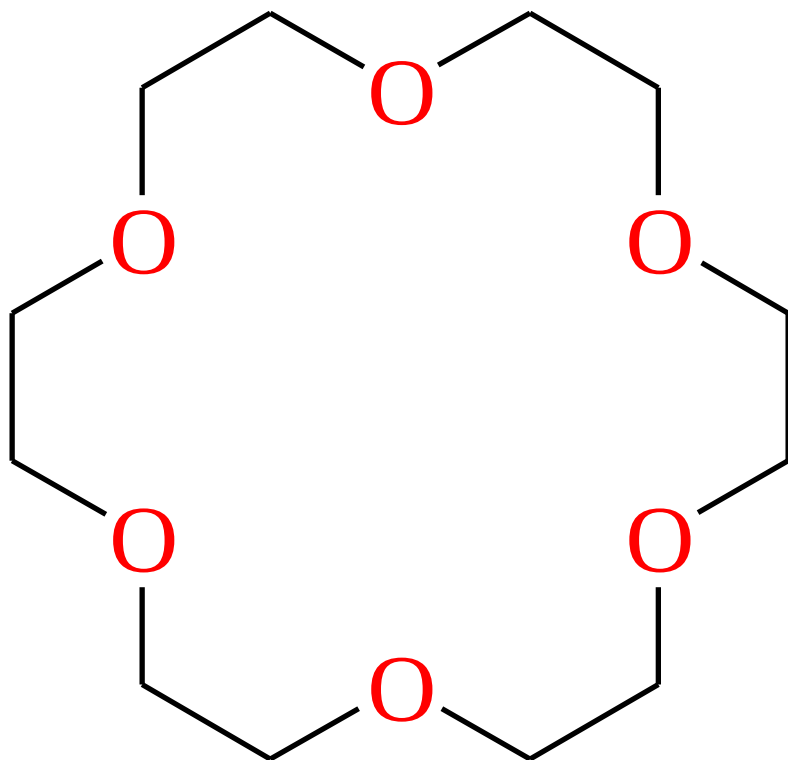


1,4-二氧六环

大环多醚

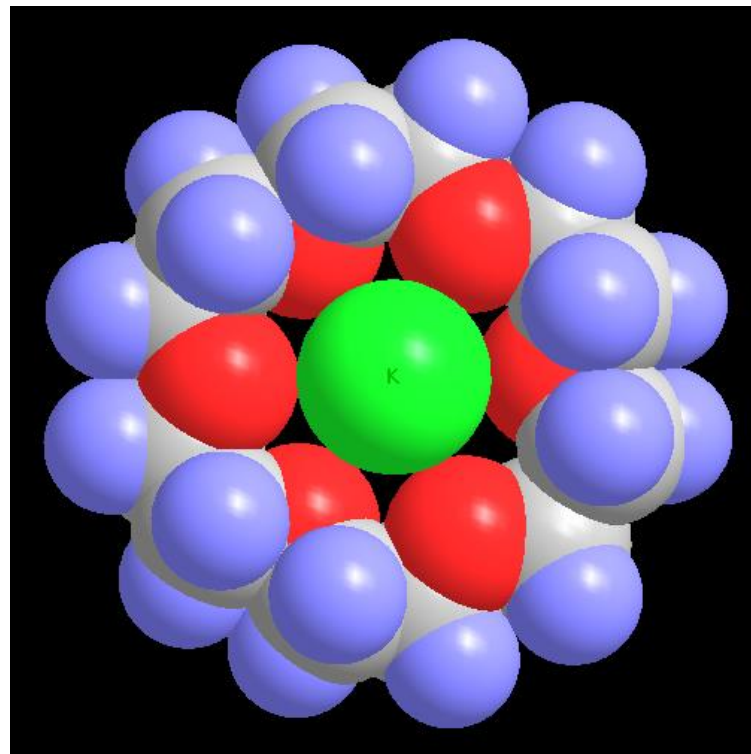
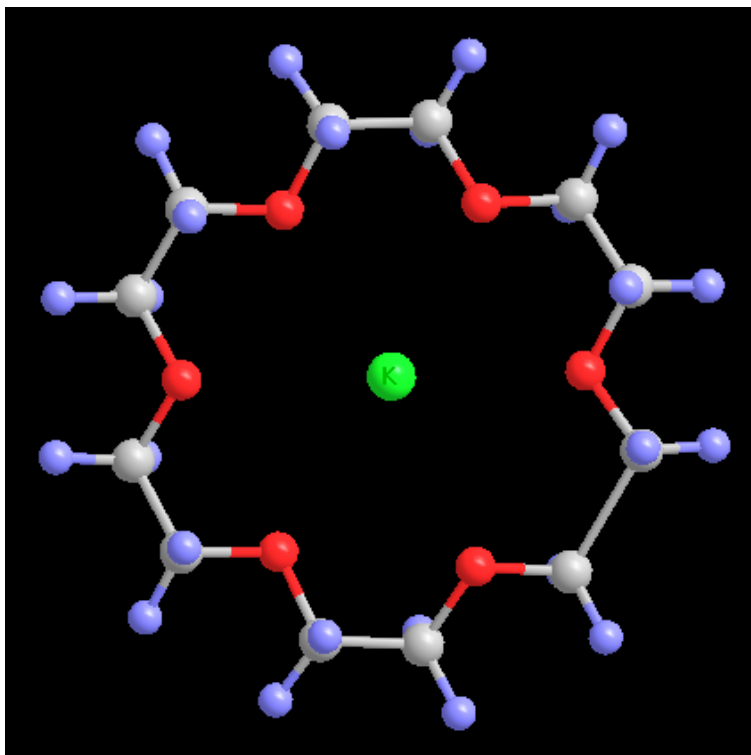
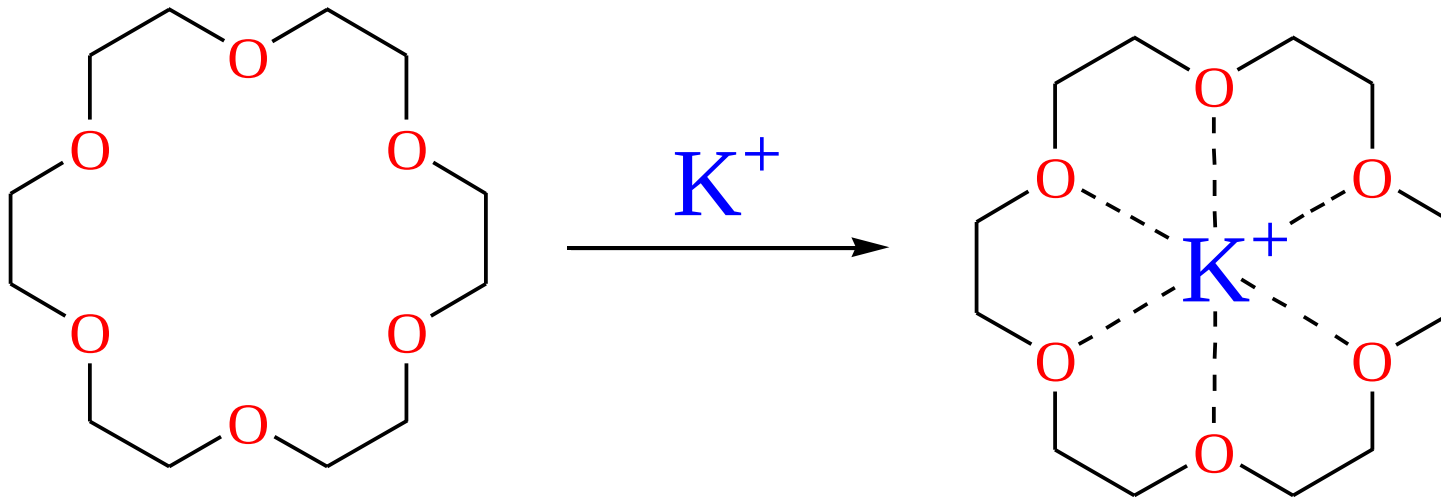


冠醚(Crown ether)



18—冠—6

18—crown—6



醚的物理性质

- b.p. 比同碳原子数的醇低得多
- 水中溶解度小(环醚除外)



b.p.(°C) -24.9

34.6

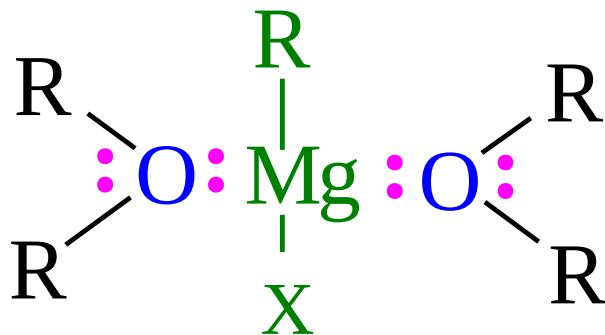
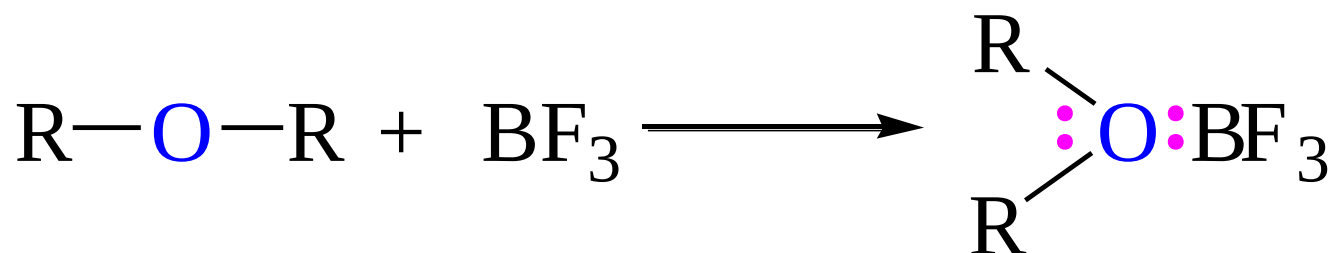
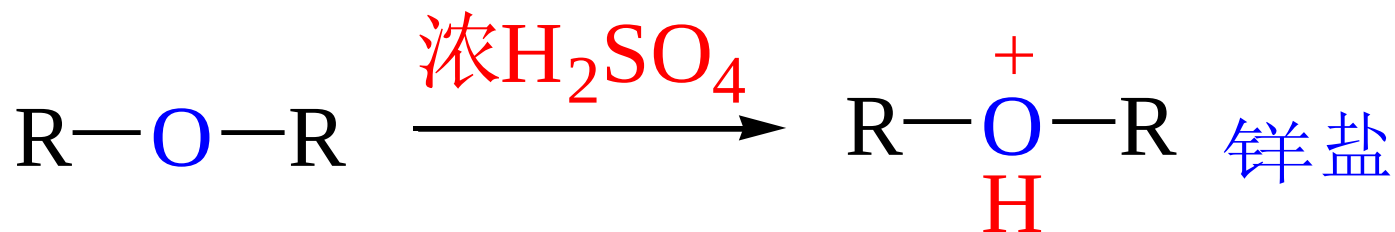


b.p.(°C) 78.4

117.8

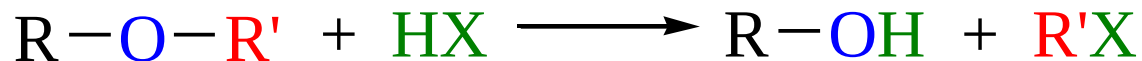
醚的化学反应

1. 形成 铕 盐

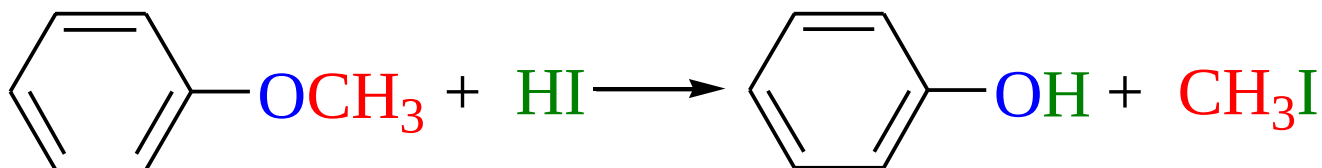
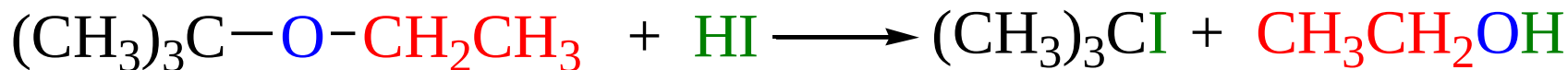
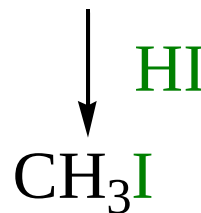
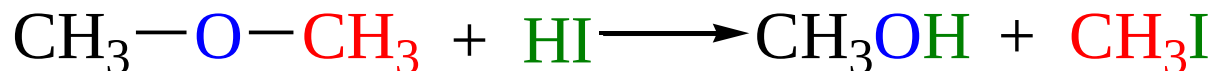


醚的化学反应

2. 醚键的断裂

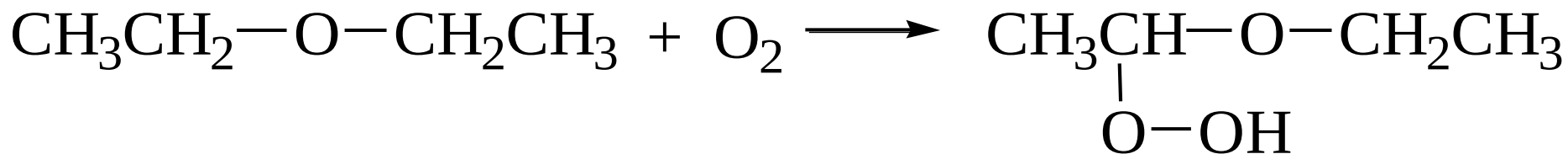


反应活性 $\text{HI} > \text{HBr} > \text{HCl}$



醚的化学反应

3. 过氧化物peroxide的生成

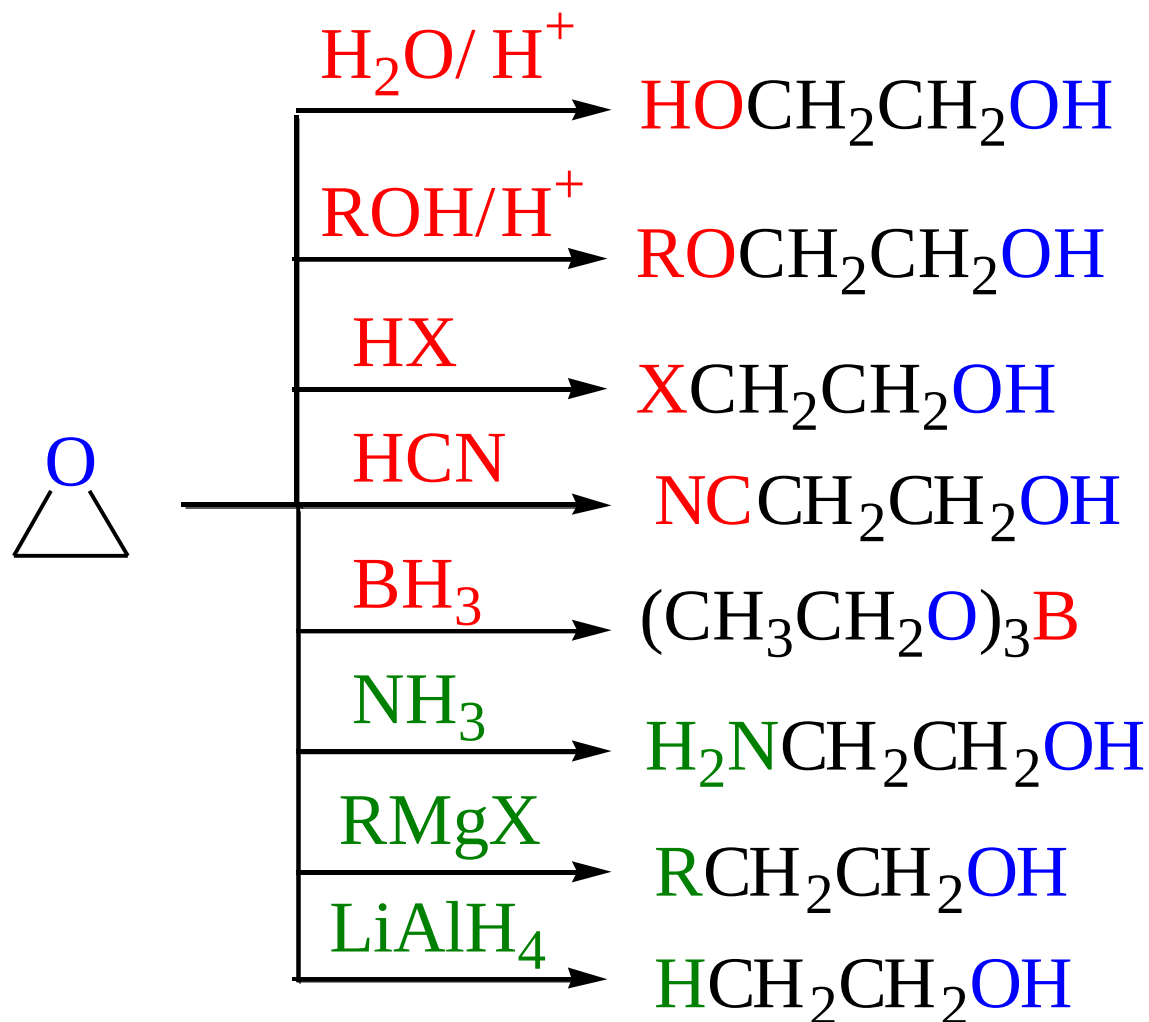


氢过氧化乙醚

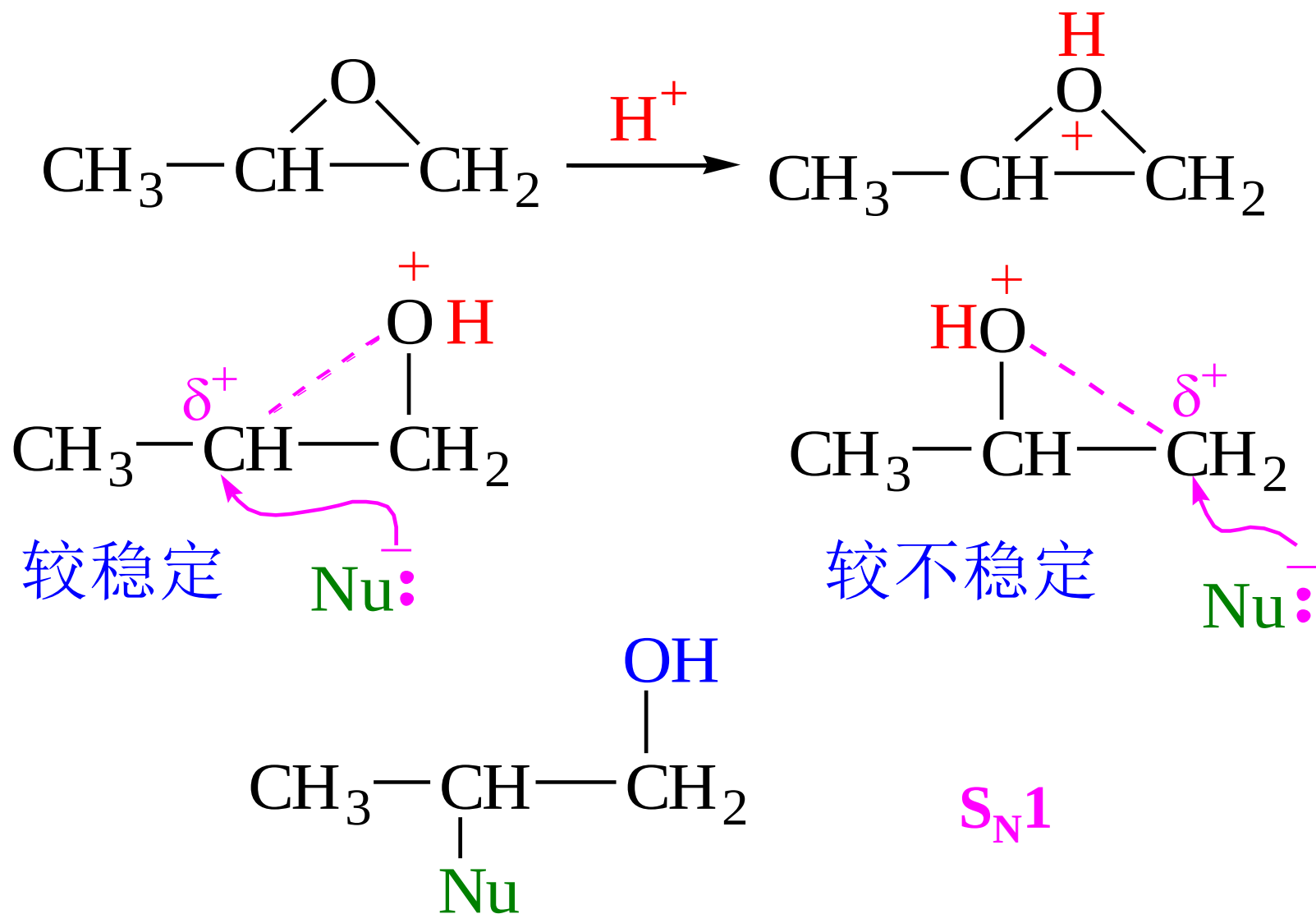
- 检验：淀粉—碘化钾试纸
- 除去过氧化物的方法：FeSO₄洗涤
- 储存时加Fe可防止形成过氧化物

醚的化学反应

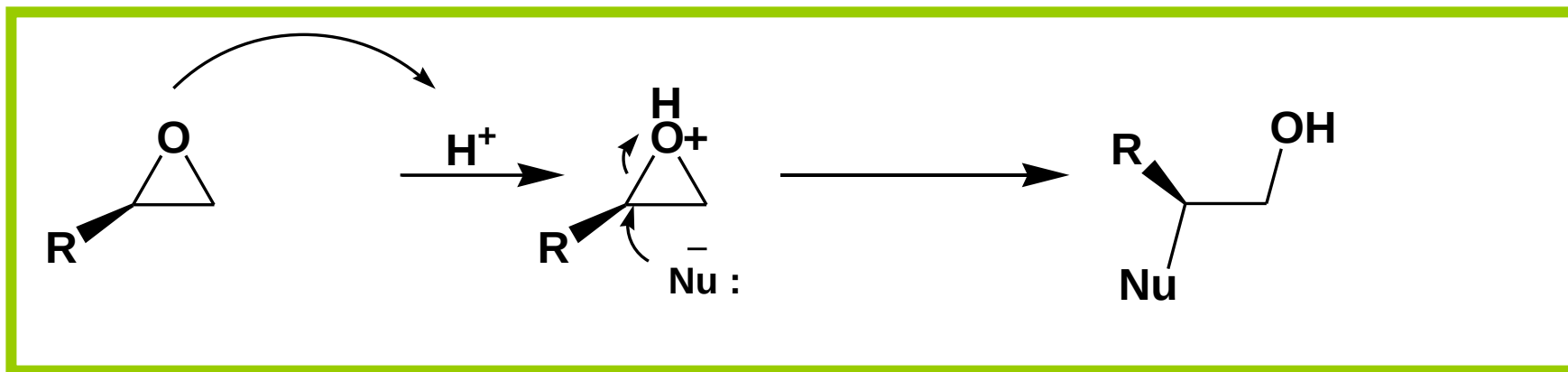
4. 环氧化合物的反应



酸催化下的开环反应



酸催化下的开环反应



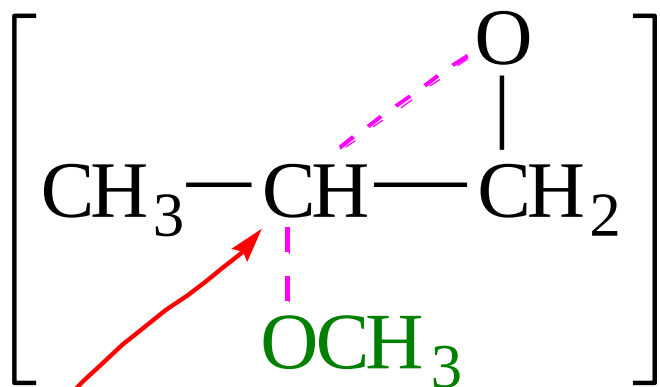
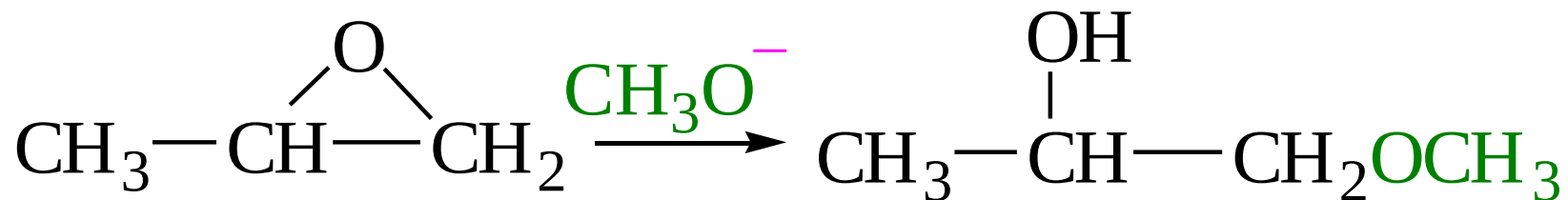
✓ H_2O

✓ ROH

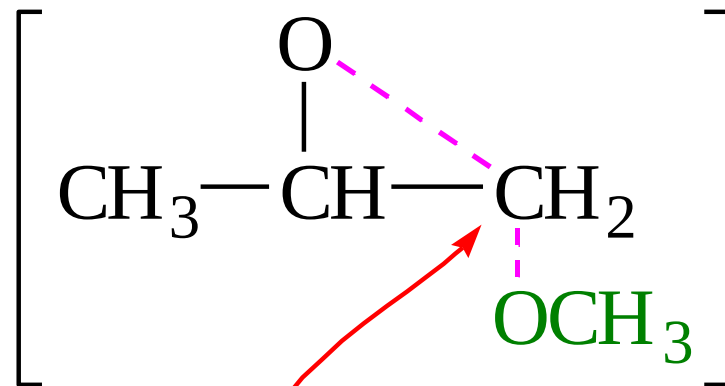
✓ HX

✓ HCN

碱催化下的开环反应



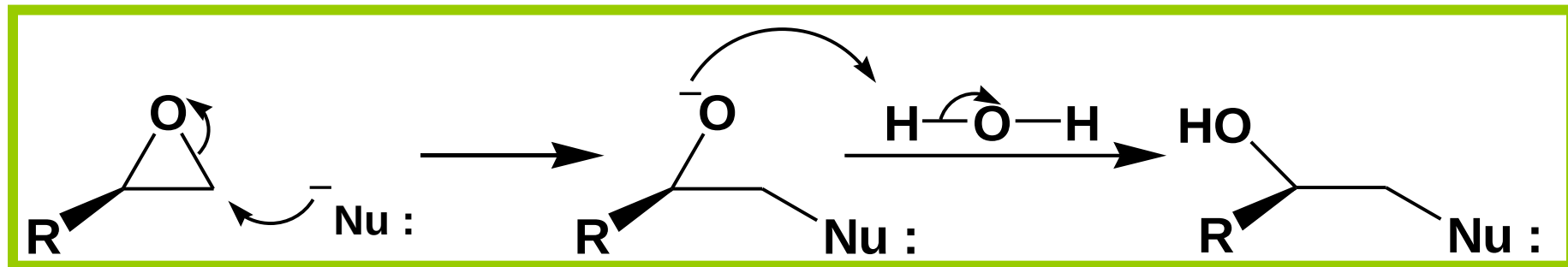
位阻较大



位阻较小

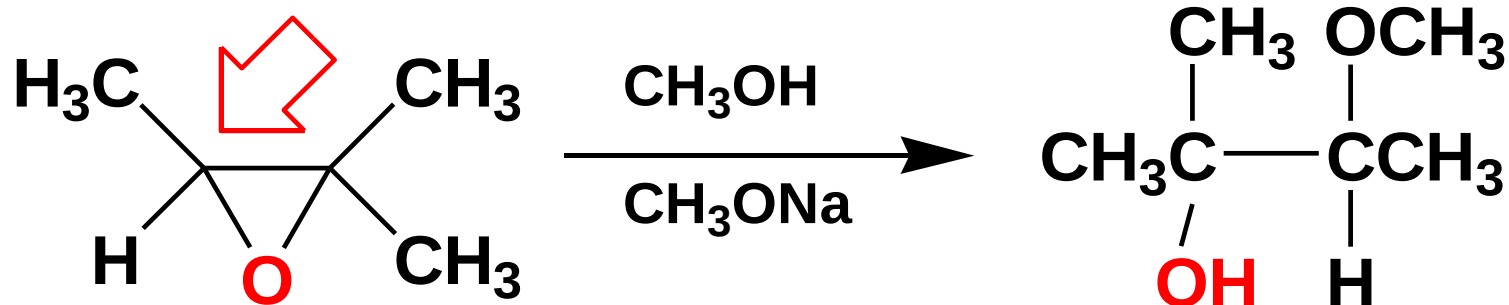
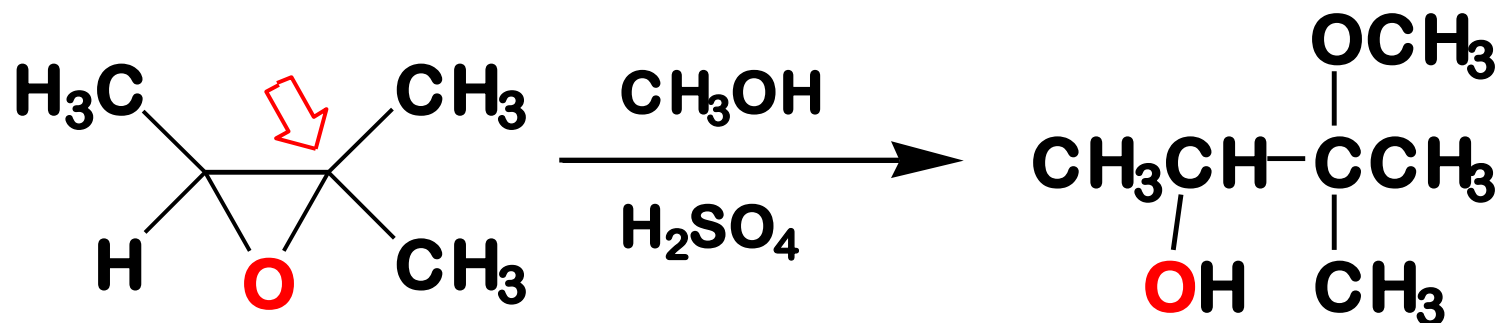
$\text{S}_{\text{N}}2$

碱催化下的开环反应

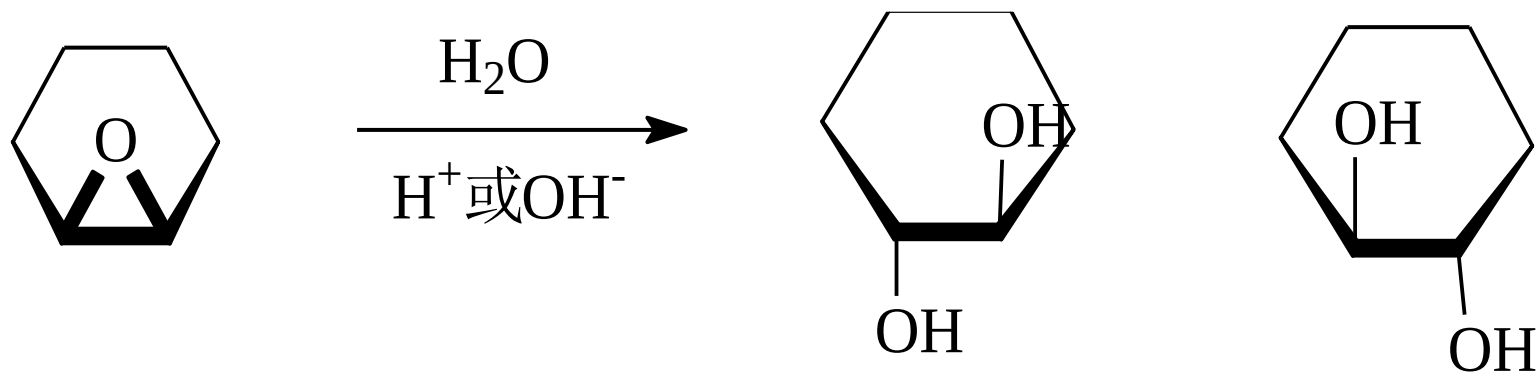
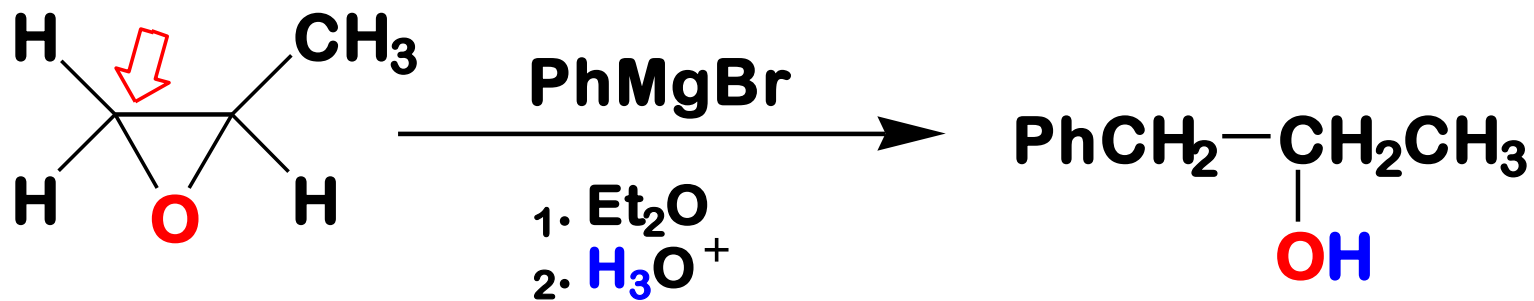


- RMgX
- RLi
- $\text{RC}\equiv\text{CNa}$
- HO^-
- RO^-

Example

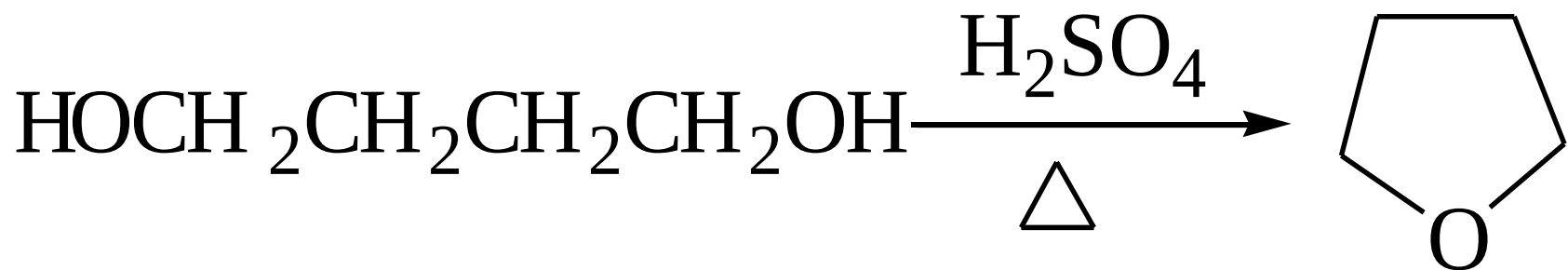
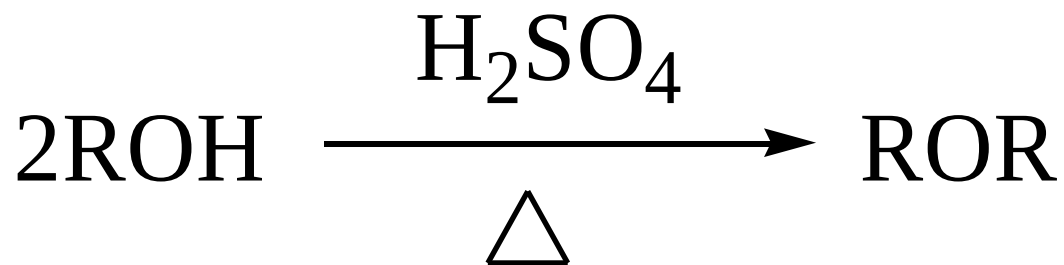


Example



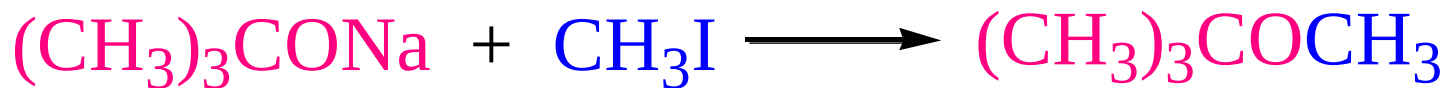
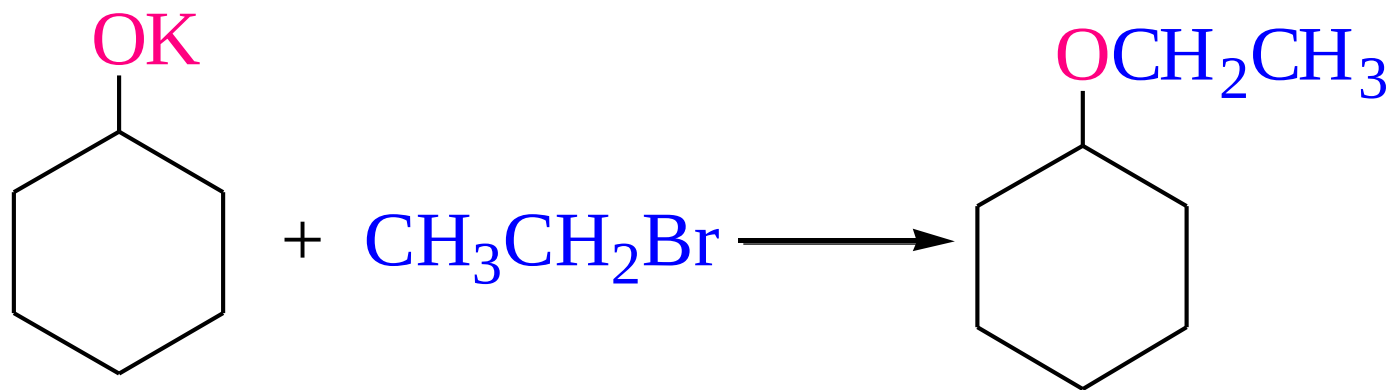
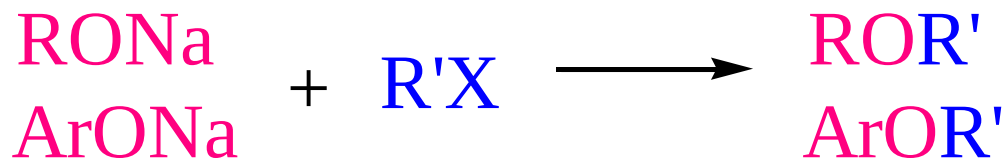
醚的制备

1. 醇的分子间/内脱水



醚的制备

2. Williamson 合成法



测试

1、下列两个制备甲基叔丁基醚的反应，哪个产率高？



2、用环己烯制备如下化合物

